

LPDDR5X SDRAM

MT62F1G32D2, MT62F2G32D4, MT62F4G32D8,
MT62F1G64D4, MT62F2G64D8

Features

- **Architecture**
 - 21.3 GB/s maximum bandwidth per channel
 - Frequency range: 1333–5 MHz (data rate range per pin: 10,667–40 Mb/s with WCK:CK = 4:1)
 - Selectable CKR (WCK:CK = 2:1 or 4:1)
- **LPDDR5X/LPDDR5 data interface**
 - Single x16 channel/die
 - Double-data-rate command/address entry
 - Differential command clocks (CK_t/CK_c) for high-speed operation
 - Differential data clocks (WCK_t/WCK_c)
 - Optional differential read strobe (RDQS_t/RDQS_c)
 - 16n-bit or 32n-bit prefetch architecture
 - Bank Architecture: Bank Group (BG) mode, and 16-bank (16B) mode supported
 - Command-selectable burst lengths (BL = 16 or 32) in bank group or 16-bank modes
 - Background ZQ calibration/command-based ZQ calibration
 - Optional link protection (link ECC)
 - Partial-array self refresh (PASR) and partial-array auto refresh (PARC) with segment mask
- **Ultra-low-voltage core and I/O power supplies**
 - V_{DD1} = 1.70–1.95V; 1.80V TYP
 - V_{DD2H} = 1.01–1.12V; 1.05V TYP
 - V_{DD2L} = V_{DD2H} or 0.87–0.97V; 0.90V TYP
 - V_{DDQ} = 0.50V or 0.45V² TYP; 0.30V TYP (ODT off only)
- **I/O characteristics**
 - Interface-LVSTL 0.5/0.3
 - I/O type: Low-swing single-ended, V_{SS} terminated
 - V_{OH} -compensated output drive
 - Programmable V_{SS} on-die termination (ODT)
 - Non target ODT support
 - DVFSQ support
 - Per byte and per pin DFE support
- **Low power features**
 - Enhanced DVFSQ: Enhanced dynamic voltage frequency scaling core
 - Single-ended CK, single-ended WCK and single-ended RDQS
 - Data copy
 - Write X

Options

- **Operating Voltage**
 - $V_{DD1}/V_{DD2H}/V_{DD2L}/V_{DDQ}/V_{DDQ}$ (ODT off only): 1.80V/1.05V/ V_{DD2H} or 0.90V/0.50V or 0.45V¹/0.30V
- **Array Configuration**
 - 1Gb x 32 (1Gbx16 x 1 die x 2 Ch x 1R)
 - 2Gb x 32 (1Gbx16 x 1 die x 2 Ch x 2R)
 - 4Gb x 32 (2Gbx8 x 2 die x 2 Ch x 2R)
 - 1Gb x 64 (1Gbx16 x 1 die x 4 Ch x 1R)
 - 2Gb x 64 (1Gbx16 x 1 die x 4 Ch x 2R)
- **Device configuration**
 - 2 die in package (1Gb16 x 2 die)
 - 4 die in package (1Gb16 x 4 die)
 - 8 die in package (2Gb8 x 8 die, 1Gb16 x 8 die)
- **FBGA RoHS-compliant “green” package**
 - 315-ball TFBGA: 12.4mm x 15.0mm (TYP), seated height 1.1mm (MAX)
 - 315-ball LFBGA: 12.4mm x 15.0mm (TYP), seated height 1.3mm (MAX)
 - 441-ball TFBGA: 14.0mm x 14.0mm (TYP), seated height 1.1mm (MAX)
 - 561-ball TFBGA: 8.0mm x 12.4mm (TYP), seated height 1.2mm (MAX)
- **Speed grade, cycle time (^tWCK)**
 - 10,667 Mb/s¹
 - 9600 Mb/s
- **Operating Temperature:**
 - -25°C ≤ T_C ≤ +85°C
 - -40°C ≤ T_C ≤ +95°C
- **Revision**

Marking

F

1G32

2G32

4G32

1G64

2G64

D2

D4

D8

DS

DV

EK

CZ

-018

-020

WT

IT

:D

Notes: 1. Supports 10,667 Mb/s (Micron specification). Refer to 10,667 Mb/s LPDDR5X Addendum.
2. V_{DDQ} = 0.45V (TYP) support on all packages up to 7500 Mb/s.

Part Number Ordering Information

Figure 1: Part Number Chart

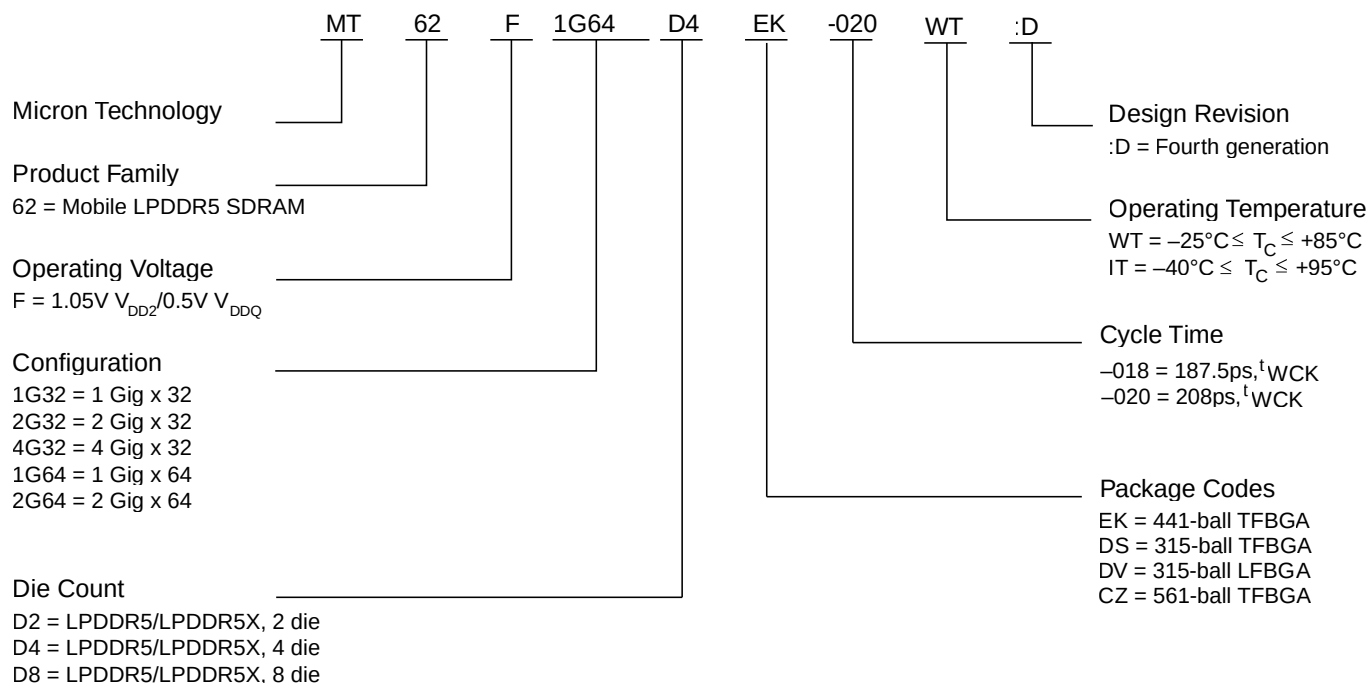


Table 1: Part Number List

Part Number	Total Density	Data Rate per Pin
MT62F1G32D2DS-020 WT:D	4GB (32Gb)	9600 Mb/s
MT62F1G32D2DS-020 IT:D		
MT62F2G32D4DS-020 WT:D	8GB (64Gb)	
MT62F2G32D4DS-020 IT:D		
MT62F4G32D8DV-020 WT:D	16GB (128Gb)	
MT62F4G32D8DV-020 IT:D		
MT62F1G64D4EK-020 WT:D	8GB (64Gb)	
MT62F2G64D8EK-020 WT:D	16GB (128Gb)	
MT62F1G64D4CZ-020 WT:D	8GB (64Gb)	
MT62F2G64D8CZ-020 WT:D	16GB (128Gb)	

Table 2: Micron 10,667 Mb/s Part Number List

Part Number	Total Density	Data Rate per Pin
MT62F1G32D2DS-018 WT:D	4GB (32Gb)	10,667 Mb/s ¹
MT62F2G32D4DS-018 WT:D	8GB (64Gb)	
MT62F4G32D8DV-018 WT:D	16GB (128Gb)	

Note: 1. Supports 10,667 Mb/s Micron specification. Refer to 10,667 Mb/s LPDDR5X Addendum.

FBGA Part Marking Decoder

Due to space limitations, FBGA-packaged components have an abbreviated part marking that is different from the part number. Micron's FBGA part marking decoder is available at www.micron.com/decoder.

LPDDR5/LPDDR5X Data Sheet List

For general LPDDR5/LPDDR5X specifications, please refer to the data sheets below.

- General LPDDR5/LPDDR5X Specifications 1: Mode Registers
- General LPDDR5/LPDDR5X Specifications 2: AC/DC and Interface Specifications
- General LPDDR5/LPDDR5X Specifications 3: Features and Functionalities

Contents

Features.....	1
Part Number Ordering Information.....	2
FBGA Part Marking Decoder.....	3
LPDDR5/LPDDR5X Data Sheet List.....	3
Contents.....	4
List of Tables.....	6
List of Figures.....	5
Important Notes and Warnings.....	7
General Notes.....	8
Device Configuration (x16, x32, x64).....	9
Refresh Requirement Parameters.....	11
CS Rx Relaxed Specification.....	11
DQ Rx Mask and Single Input Pulse Relaxed Specifications.....	12
DQ Rx Mask and Single Input Pulse Relaxed Specifications 3200 MHz.....	14
16-Bank Extended Frequency.....	15
16-Bank Extended Frequency Description.....	15
16B/BG Data Rate Change Diagram With FSP Requirements.....	16
Package Block Diagrams.....	17
Dual Die, Dual Channel, Single Rank.....	17
Quad Die, Dual Channel, Dual Rank.....	18
Eight Die, Dual Channel, Dual Rank.....	19
Quad Die, Quad Channel, Single Rank.....	20
Eight Die, Quad Channel, Dual Rank.....	21
Ball Assignments and Descriptions.....	22
315b Single Die, Single Channel, 1 Rank, 2 Rank.....	22
315b Dual Channel, 1 Rank, 2 Rank.....	24
441-Ball Quad Channel, 1 Rank, 2 Rank.....	26
561-Ball Quad Channel, 1 Rank, 2 Rank.....	28
Package Dimensions.....	30
315-Ball Package (Package Code: DS).....	30
315-Ball Package (Package Code: DV).....	31
441-Ball Package (Package Code: EK).....	32
561-Ball Package (Package Code: CZ).....	33
Package Thermal Impedance.....	34
Product-Specific Mode Register Definition.....	35
I _{DD} Parameters.....	37
Revision History.....	45
Rev. G – 03/2026.....	45
Rev. F – 02/2026.....	45
Rev. E – 01/2026.....	45
Rev. D – 12/2025.....	45
Rev. C – 08/2025.....	45
Rev. B – 06/2025.....	45
Rev. A – 06/2024.....	45

List of Figures

Figure 1: Part Number Chart.....	2
Figure 2: DQ Rx Mask Definition.....	13
Figure 3: LPDDR5X DQ Single Pulse Definition.....	13
Figure 4: 16B and BG Data Rate Change Flow Chart Showing Extended Capabilities.....	16
Figure 5: Dual-Die, Dual-Channel, Single-Rank Package Block Diagram.....	17
Figure 6: Quad-Die, Dual-Channel, Dual-Rank Package Block Diagram.....	18
Figure 7: Eight-Die, Dual-Channel, Dual-Rank Package Block Diagram.....	19
Figure 8: Quad-Die, Quad-Channel, Single-Rank Package Block Diagram.....	20
Figure 9: Eight-Die, Quad-Channel, Dual-Rank Package Block Diagram.....	21
Figure 10: 315-Ball Single-Channel Discrete FBGA.....	23
Figure 11: 315-Ball Dual-Channel Discrete FBGA.....	25
Figure 12: 441-Ball Quad-Channel FBGA.....	27
Figure 13: 561-Ball Quad-Channel FBGA.....	29
Figure 14: 315-Ball TFBGA – 12.4mm (TYP) × 15.0mm (TYP) × 1.1mm (MAX).....	30
Figure 15: 315-Ball LFBGA – 12.4mm (TYP) × 15.0mm (TYP) × 1.3mm (MAX).....	31
Figure 16: 441-Ball TFBGA – 14.0mm (TYP) × 14.0mm (TYP) × 1.1mm (MAX).....	32
Figure 17: 561-Ball TFBGA – 8.0mm (TYP) × 12.4mm (TYP) × 1.2mm (MAX).....	33

List of Tables

Table 1: Part Number List.....	2
Table 2: Micron 10,667 Mb/s Part Number List.....	2
Table 3: Die Organization in the Package (x32).....	9
Table 4: Die Organization in the Package (x64).....	9
Table 5: Die Addressing.....	10
Table 6: Refresh Requirement Parameters.....	11
Table 7: CS Rx Specifications.....	11
Table 8: DQ Rx Mask and Single Input Pulse Specifications.....	12
Table 9: DQ Rx Mask and Single Input Pulse Specifications for 10.7 Gb/s.....	12
Table 10: DQ, DMI, Parity and DBI Rx Mask and Single Input Pulse Specifications.....	14
Table 11: 16-Bank Operation ¹ @ 4.3 Gb/s.....	15
Table 12: Single Channel 315-Ball/Pad Descriptions.....	22
Table 13: 315-Ball/Pad Descriptions.....	24
Table 14: 441-Ball/Pad Descriptions.....	26
Table 15: Ball/Pad Descriptions.....	28
Table 16: Package Thermal Impedance Characteristics.....	34
Table 17: Mode Register Contents.....	35
Table 18: WT I _{DD} Parameters – Single Die.....	37
Table 19: WT I _{DD} Parameters – with Enhanced DVFSC enabled - Single Die.....	39
Table 20: WT Full-Array Power-Down Self Refresh Current – Single Die.....	40
Table 21: IT I _{DD} Parameters – Single Die.....	41
Table 22: IT I _{DD} Parameters – with Enhanced DVFSC enabled - Single Die.....	43
Table 23: IT Full-Array Power-Down Self Refresh Current – Single Die.....	44

Important Notes and Warnings

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General Notes

Throughout the data sheet, figures and text refer to DQs as DQ. DQ should be interpreted as any or all DQs collectively, unless specifically stated otherwise.

RDQS, CK, and WCK should be interpreted as RDQS_t, RDQS_c, CK_t, CK_c, and WCK_t, WCK_c respectively unless specifically stated otherwise. CA includes all CA pins used for a given density.

In timing diagrams, CMD is used as an indicator only. Actual signals occur on CA[6:0].

V_{REF} indicates $V_{REF(CA)}$, $V_{REF(CS)}$, and $V_{REF(DQ)}$.

Complete functionality is described throughout the entire document. Any page or diagram may have been simplified to convey a topic and may not be inclusive of all requirements.

Any content defined in this device ID specific data sheet document takes precedence over similar content defined in the general core LPDDR5/LPDDR5X SDRAM data sheet document(s).

Any functionality not specifically stated herein is considered undefined, illegal, is not supported, and will result in unknown operation.

Device Configuration (x16, x32, x64)

Table 3: Die Organization in the Package (x32)

Die Organization	1G32 (32 Gb/package)	2G32 (64 Gb/package)	4G32 (128 Gb/package)
Channel A	x16 mode × 1 die	–	–
Channel B	x16 mode × 1 die	–	–
Channel A, rank 0	–	x16 mode × 1 die	–
Channel B, rank 0	–	x16 mode × 1 die	–
Channel A, rank 1	–	x16 mode × 1 die	–
Channel B, rank 1	–	x16 mode × 1 die	–
Channel A, rank 0 DQ[7:0]	–	–	x8 mode × 1 die
Channel A, rank 1 DQ[7:0]	–	–	x8 mode × 1 die
Channel B, rank 0 DQ[7:0]	–	–	x8 mode × 1 die
Channel B, rank 1 DQ[7:0]	–	–	x8 mode × 1 die
Channel A, rank 0 DQ[15:8]	–	–	x8 mode × 1 die
Channel A, rank 1 DQ[15:8]	–	–	x8 mode × 1 die
Channel B, rank 0 DQ[15:8]	–	–	x8 mode × 1 die
Channel B, rank 1 DQ[15:8]	–	–	x8 mode × 1 die

Note: 1. Refer to the Package Block Diagram section in this data sheet.

Table 4: Die Organization in the Package (x64)

Die Organization	1G64 (64 Gb/package)	2G64 (128 Gb/package)
Channel A, rank 0	x16 mode × 1 die	x16 mode × 1 die
Channel B, rank 0	x16 mode × 1 die	x16 mode × 1 die
Channel C, rank 0	x16 mode × 1 die	x16 mode × 1 die
Channel D, rank 0	x16 mode × 1 die	x16 mode × 1 die
Channel A, rank 1	–	x16 mode × 1 die
Channel B, rank 1	–	x16 mode × 1 die
Channel C, rank 1	–	x16 mode × 1 die
Channel D, rank 1	–	x16 mode × 1 die

Note: 1. Refer to the Package Block Diagram section in this data sheet.

Table 5: Die Addressing

Description	1G32 (32 Gb/package), 2G32 (64 Gb/package), 1G64 (64 Gb/package), 2G64 (128 Gb/package)		4G32 (128 Gb/package)	
	BG mode	16B mode	BG mode	16B mode
Density per die	16Gb		16Gb	
Bits	17,179,869,184		17,179,869,184	
Configuration	64Mb × 16 DQ × 4 Banks × 4BG	64Mb × 16 DQ × 16 Banks	128Mb × 8 DQ × 4 Banks × 4BG	128Mb × 8 DQ × 16 Banks
Number of banks	4	16	4	16
Number of bank groups	4	1	4	1
Array prefetch bits	256	256	128	128
Rows per bank	65,536		131,072	
Columns	64		64	
Page size (bytes)	2048	2048	1024	1024
Native burst length	16	16	16	16
Number of I/Os	16		8	
Bank address	BA[1:0]	BA[3:0]	BA[1:0]	BA[3:0]
Bank group address	BG[1:0]	–	BG[1:0]	–
Row address	R[15:0]		R[16:0]	
Column address	C[5:0]		C[5:0]	
Burst address	B[3:0]	B[3:0]	B[3:0]	B[3:0]
Burst starting address boundary	128-bit		128-bit	

Note: 1. Refer to the SDRAM Addressing section in General LPDDR5/LPDDR5X Specifications 3.

Refresh Requirement Parameters

Table 6: Refresh Requirement Parameters

Parameter	Symbol	16Gb Die		Unit
		BG and 16B Mode		
		Enhanced DVFSC Disabled	Enhanced DVFSC Enabled	
REFRESH cycle time (all banks)	^t RFCab	280	300	ns
REFRESH cycle time (per bank)	^t RFCpb	140	150	ns
Per bank refresh to per bank refresh time (different bank)	^t PBR2PBR	90	100	ns
Per bank refresh to ACTIVATE command time (different bank)	^t PBR2ACT	7.5	7.5	ns

Note: 1. This table only describes refresh parameters that are density dependent. Refer to Refresh Requirement section in General LPDDR5/LPDDR5X Specifications 3 for all refresh parameters.

CS Rx Relaxed Specification

Table 7: CS Rx Specifications

Item	Symbol	Min/ Max	CK Frequency (MHz)														Unit	Note	
			67 ⁶	133	200	266	344	400	467	533	600	688	750	800	937.5	1066.5			1200
Rx mask																			
CS Rx mask height	vCSIVW	Max	155														mV	2, 7	
CS Rx mask width at V _{REF} CS	^t CSIVW1	Max	0.3														^t CK(avg)	1	
CS Rx mask width at vCSIVW	^t CSIVW2	Max	0.22														^t CK(avg)	1	
Rx single pulse																			
CS Rx pulse amplitude	vCSIHL_AC	Min	240														mV	3	
CS reference voltage	vREFCS		V _{DD2H} /3														mV		
CS Rx pulse width	^t CSIPW	Min	0.6														^t CK(avg)		
Power down																			
CS V _{IL} during power down/deep sleep	V _{ILPD}	Max	130														mV	4	
CS V _{IH} during power down/deep sleep	V _{IHPD}	Min	550														mV	5	
		Max	V _{DD2H} + 200																

- Notes: 1. CS Rx mask voltage and timing parameters at the pin include temperature drift and voltage AC noise impact based on Z(f) specification at a fixed temperature on the package. The voltage supply noise must comply to the component min/max DC operating conditions.
2. CS single-pulse signal amplitude into the receiver must meet or exceed v_{CSIHL_AC} at any point over the total UI; No timing requirement above a certain level. v_{CSIHL_AC} is the peak-to-peak voltage centered around $V_{REF\ CS}$ such that $v_{CSIHL_AC}/2$ min has to be met both above and below $V_{REF\ CS}$.
3. v_{CSIHL_AC} does not have to be met when no transitions are occurring.
4. The input voltage presented to the CS Rx pin during power down should be 0V nominally to minimize leakage current.
5. V_{IHPD} is only applied for POWER DOWN and DEEP SLEEP EXIT operations.

6. The Rx voltage and absolute timing requirements apply for all CS operating frequencies at or below 67 for all speed bins. For example, t_{CSIVW1} (ns) = 4.477ns at or below 67 MHz CK frequency.
7. Measured at die pad.

DQ Rx Mask and Single Input Pulse Relaxed Specifications

Table 8: DQ Rx Mask and Single Input Pulse Specifications

Item	Symbol	Min/ Max	WCK Frequency (MHz)		Unit	Notes
			4266	4800		
DQ Rx mask height	vDIVW	Max	70		mV	1, 2, 3
DQ Rx mask width at $V_{REF(DQ)}$	t_{DIVW1}	Max	0.35		UI	1, 3
DQ Rx mask width at vDIVW	t_{DIVW2}	Max	0.18		UI	1, 3
DQ Rx pulse width at $V_{REF(DQ)}$	t_{DIPW1}	Min	0.51		UI	4
DQ Rx pulse reference	t_{DIPW2}	Min	0.26		UI	4
DQ Rx pulse width at $V_{REF(DQ)} \pm vDIVW/2$	t_{DIHL}	Min	0.24		UI	4
DQ Rx pulse amplitude from programmed $V_{REF(DQ)}$	vDIHP1	Min	70		mV	
	vDILP1	Max	-70		mV	
DQ Rx early pulse amplitude from programmed $V_{REF(DQ)}$	vDIHP2	Min	55		mV	
	vDILP2	Max	-55		mV	

- Notes: 1. Data Rx mask voltage and timing parameters are applied per pin and include the DRAM DQ-to-DQS voltage AC noise impact for frequencies greater than TBD MHz and maximum voltage of TBD mV peak-to-peak from DC-TBD MHz at a fixed temperature on the package. The voltage supply noise must comply with the component min/max DC operating conditions.
2. Rx mask voltage vDIVW must be centered around $V_{REF(DQ)}$.
3. Mask specifications (t_{DIVW1} , t_{DIVW2} , and vDIVW) are the same regardless DFE enabled or disabled. DFE coefficient can be included as part of input waveform amplitude when DFE is enabled.
4. UI = $t_{WCK}/2$, programed V_{REF} , is defined as a percentage of MR14/15 code x V_{DDQ} .

Table 9: DQ Rx Mask and Single Input Pulse Specifications for 10.7 Gb/s

Item	Symbol	Min/ Max	WCK Frequency (MHz)		Unit	Notes
			5333			
DQ Rx mask height	vDIVW	Max	70		mV	1, 2, 3
DQ Rx mask width at $V_{REF(DQ)}$	t_{DIVW1}	Max	36.5		ps	1, 3
DQ Rx mask width at vDIVW	t_{DIVW2}	Max	18.8		ps	1, 3
DQ Rx pulse width at $V_{REF(DQ)}$	t_{DIPW1}	Min	46.8		ps	4
DQ Rx pulse reference	t_{DIPW2}	Min	27.0		ps	4
DQ Rx pulse width at $V_{REF(DQ)} \pm vDIVW/2$	t_{DIHL}	Min	21.8		UI	4
DQ Rx pulse amplitude from programmed $V_{REF(DQ)}$	vDIHP1	Min	70		mV	
	vDILP1	Max	-70		mV	
DQ Rx early pulse amplitude from programmed $V_{REF(DQ)}$	vDIHP2	Min	55		mV	
	vDILP2	Max	-55		mV	

- Notes: 1. Data Rx mask voltage and timing parameters are applied per pin and include the DRAM DQ-to-DQS voltage AC noise impact for frequencies greater than TBD MHz and maximum voltage of TBD mV peak-to-peak from DC-TBD MHz at a fixed temperature on the package. The voltage supply noise must comply with the component min/max DC operating conditions.
2. Rx mask voltage vDIVW must be centered around $V_{REF(DQ)}$.

3. Mask specifications (t_{DIVW1} , t_{DIVW2} , and v_{DIVW}) are the same regardless DFE enabled or disabled. DFE coefficient can be included as part of input waveform amplitude when DFE is enabled.
4. $UI = t_{WCK}/2$, programed V_{REF} , is defined as a percentage of MR14/15 code $\times V_{DDQ}$.

Figure 2: DQ Rx Mask Definition

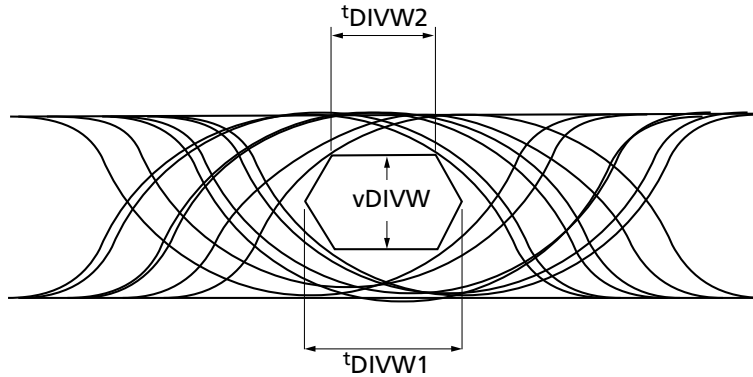
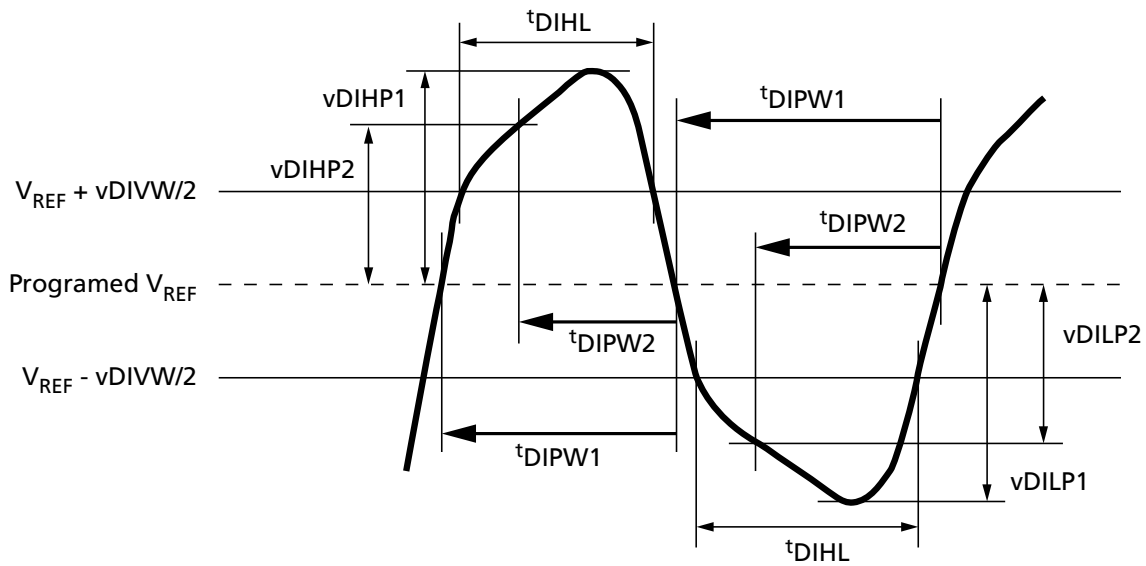


Figure 3: LPDDR5X DQ Single Pulse Definition



Note: 1. $v_{DIHL_AC}/2 = v_{DIHP1} = v_{DILP1}$. Programed $V_{REF(DQ)}$ is defined as percentage of MR14/15 code $\times V_{DDQ}$.

DQ Rx Mask and Single Input Pulse Relaxed Specifications 3200 MHz

Table 10: DQ, DMI, Parity and DBI Rx Mask and Single Input Pulse Specifications

Item	Symbol	Min/Max	WCK Frequency (MHz)	Unit	Notes
			3200		
DQ Rx mask height	vDIVW	Max	70	mV	1, 2, 3, 6
DQ Rx mask width at $V_{REF(DQ)}$	t^{DIVW1}	Max	0.35	UI	1, 3
DQ Rx mask width at vDIVW	t^{DIVW2}	Max	0.18	UI	1, 3
DQ Rx pulse width at $V_{REF(DQ)}$	t^{DIPW1}	Min	0.51	UI	4, 6
DQ Rx pulse reference	t^{DIPW2}	Min	0.26	UI	4
DQ Rx pulse width at $V_{REF(DQ)} \pm vDIVW/2$	t^{DIHL}	Min	0.24	UI	4, 6
DQ Rx pulse amplitude from programmed $V_{REF(DQ)}$	vDIHP1	Min	70	mV	
	vDILP1	Max	-70	mV	
DQ Rx early pulse amplitude from programmed $V_{REF(DQ)}$	vDIHP2	Min	55	mV	
	vDILP2	Max	-55	mV	
DQ V_{REF}	$V_{REF(DQ)}$	Min	75	mV	6
		Max	180	mV	6
DQ to DQ offset	t^{DQ2DQ}	Max	30	ps	5

- Notes: 1. Data Rx mask voltage and timing parameters are applied per pin and include the DRAM DQ-to-DQS voltage AC noise impact for frequencies greater than TBD MHz and maximum voltage of TBD mV peak-to-peak from DC-TBD MHz at a fixed temperature on the package. The voltage supply noise must comply with the component min/max DC operating conditions.
2. Rx mask voltage vDIVW must be centered around $V_{REF(DQ)}$.
3. Mask specifications (t^{DIVW1} , t^{DIVW2} , and vDIVW) are the same regardless DFE enabled or disabled. DFE co-efficient can be included as part of input waveform amplitude when DFE is enabled.
4. UI = $t^{WCK}/2$, programed V_{REF} , is defined as a percentage of MR14/15 code x V_{DDQ} .
5. DQ to DQ offset defined within byte from DRAM pin to DRAM internal latch for a given component.
6. When vDIVW is 70mV or more and less than 80mV.

16-Bank Extended Frequency

16-Bank Extended Frequency Description

Micron's LPDDR5 SDRAM device which are marked with 8533 Mb/s per-pin (-023) capability can support frequency set points using 2:1 WCK to CK ratio and above the 1600 MHz limit. Shown in the table below is specification requirements when considering 1867 MHz and 2133 MHz frequencies. This support is not available on 1-alpha process node and earlier Device ID's, this feature can expand capabilities of your system.

Table 11: 16-Bank Operation¹ @ 4.3 Gb/s

Parameter	3733 Mb/s per-pin	Comments	4267 Mb/s per-pin	Comments
RL Set 1/2/3	20/22/24	MR2[3:0] = 1100	23/25/26	MR2[3:0] = 1101
nRBTP	6		6	
WL Set A/B	11/19	MR1[7:4] = 1100	12/22	MR1[7:4] = 1101
nWR x16/x8	32/34	MR2[7:4] = 1100	37/39	MR2[7:4] = 1101
^t WCKPRE_static ²	5	Follows 7500 data rate	6	Follows 8533 data rate
^t WCKPRE_WR_toggle	4	Follows 3200 data rate	4	Follows 3200 data rate
^t WCKPRE_RD_toggle	10	Follows 3200 data rate	10	Follows 3200 data rate
^t WCKENL_FS	2	Follows 3200 data rate	2	Follows 3200 data rate
^t WCKPRE_toggle_FS ³	14	Minimum	16	Minimum
ODTLon	WL - 4	Follows 3200 data rate	WL - 4	Follows 3200 data rate
ODTLoFF ⁴	WL + 5 (BL16), WL + 9 (BL32)	Follows 3200	WL + 5 (BL16), WL + 9 (BL32)	Follows 3200 data rate
ODTLoFF_RD_DQ	RL - 4	Follows 3200 data rate	RL - 4	Follows 3200 data rate
ODTLoFF_RD_RDQS	RL - 6	MR10 OP[1, 5:4] = 000, 001, 010	RL - 6	MR10 OP[1, 5:4] = 000, 001, 010
	RL - 8	MR10 OP[1, 5:4] = 011, 1XX	RL - 8	MR10 OP[1, 5:4] = 011, 1XX
ODTLon_RD_DQ ⁵	RL + 6 (BL16), RL + 10 (BL32)		RL + 6 (BL16), RL + 10 (BL32)	
ODTLon_RD_RDQS ⁶	RL + 8 (BL16), RL + 12 (BL32)		RL + 8 (BL16), RL + 12 (BL32)	

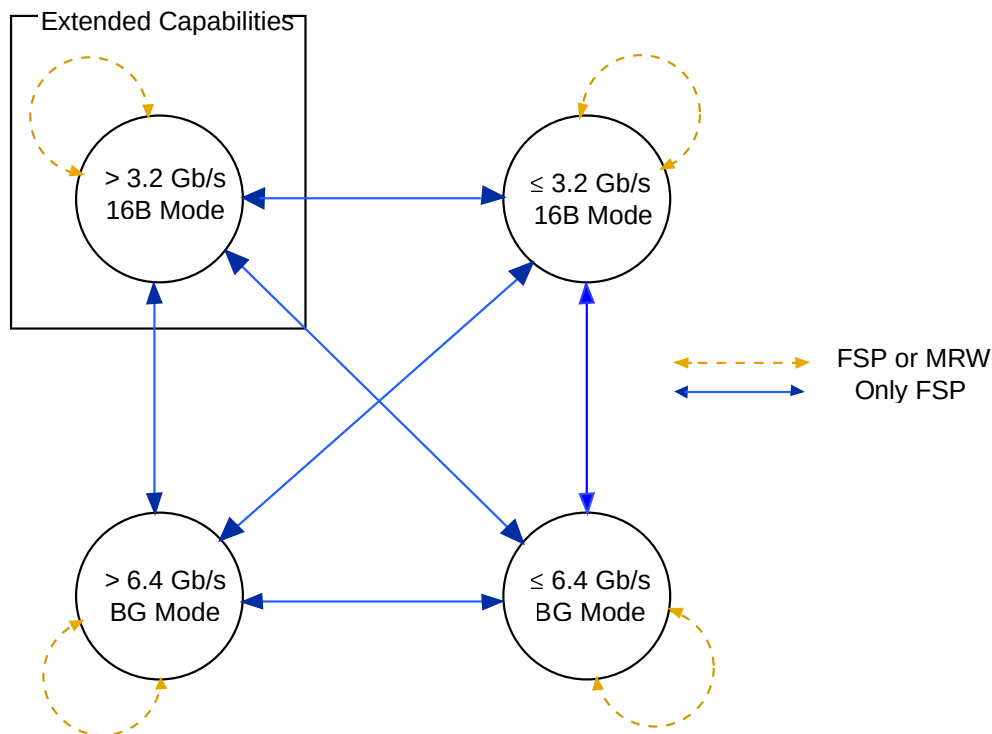
- Notes: 1. Restricted to WCK:CK 2:1 ratio, WCK:CK 4:1 ratio not supported; Link ECC is disabled, and DVFS/E-DVFS is disabled.
2. Common for WR/RD/FS.
3. Minimum 1 + RL - (^tWCKENL_FS + ^tWCKPRE_Static) when CAS to RD is 1CK.
4. (WL + BL/n_min + RU(^tWCK2DQI(max)/^tCK)).
5. (RL + BL/n_min + RU(^tWCK2DQO(max)/^tCK)).
6. (RL + BL/n_min + RU(^tWCK2DQO(max)/^tCK)) + 2.
7. 8533 MT/s or higher data rate parts support 16-bank mode up to 4267 MT/s with restrictions, all other speed rated parts greater than 3200 MT/s per-pin and less than 8533 MT/s per-pin are restricted from the extended 16B mode capabilities. Refer to the above table for constraints for operating in 16-bank mode when data rate is greater than 3200 MT/s per-pin and equal to or less than 4267 MT/s per-pin.
8. When operating in the 16-bank mode, the FSP procedure must be used when setting the data rate above 3.2 Gb/s per-pin even when setting for the first time. For example, if the device is powered-up and initialized for 4266 Gb/s per-pin timing at the initial operation start, the FSP procedure must still be used. See [Figure 4: 16B and BG Data Rate Change Flow Chart Showing Extended Capabilities](#) on page 16 for details of FSP usage.

16B/BG Data Rate Change Diagram With FSP Requirements

The following flow chart diagram shows the needed 16-bank (16B) and bank group (BG) data rate change details and how the frequency set protocol (FSP) supports the extended frequency range.

The diagram below represents the device extended capabilities which are extended to 4267 Mb/s per-pin data rate in 16B mode.

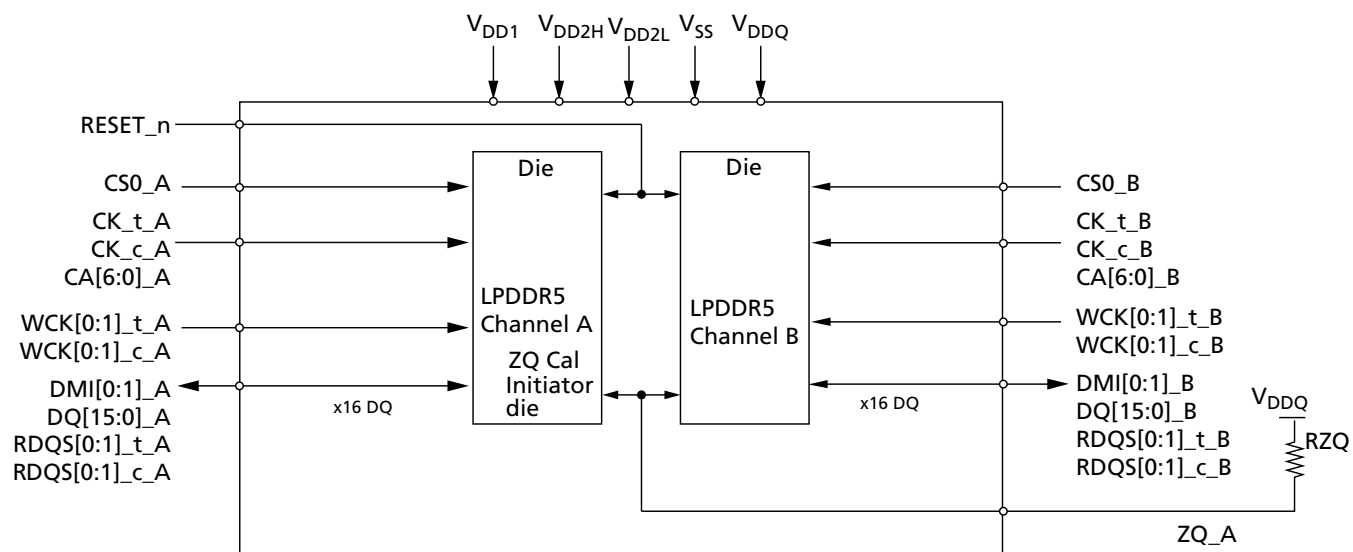
Figure 4: 16B and BG Data Rate Change Flow Chart Showing Extended Capabilities



Package Block Diagrams

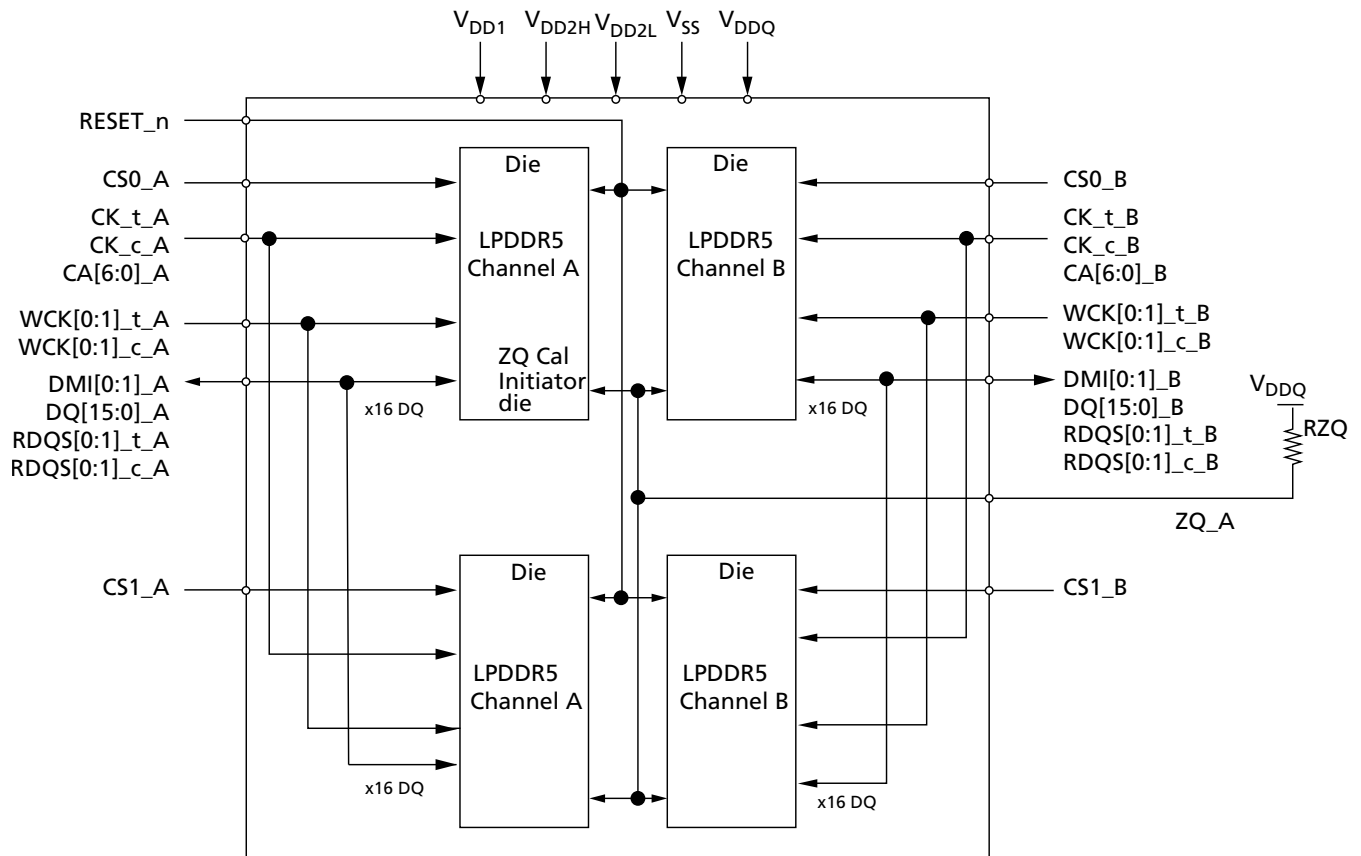
Dual Die, Dual Channel, Single Rank

Figure 5: Dual-Die, Dual-Channel, Single-Rank Package Block Diagram



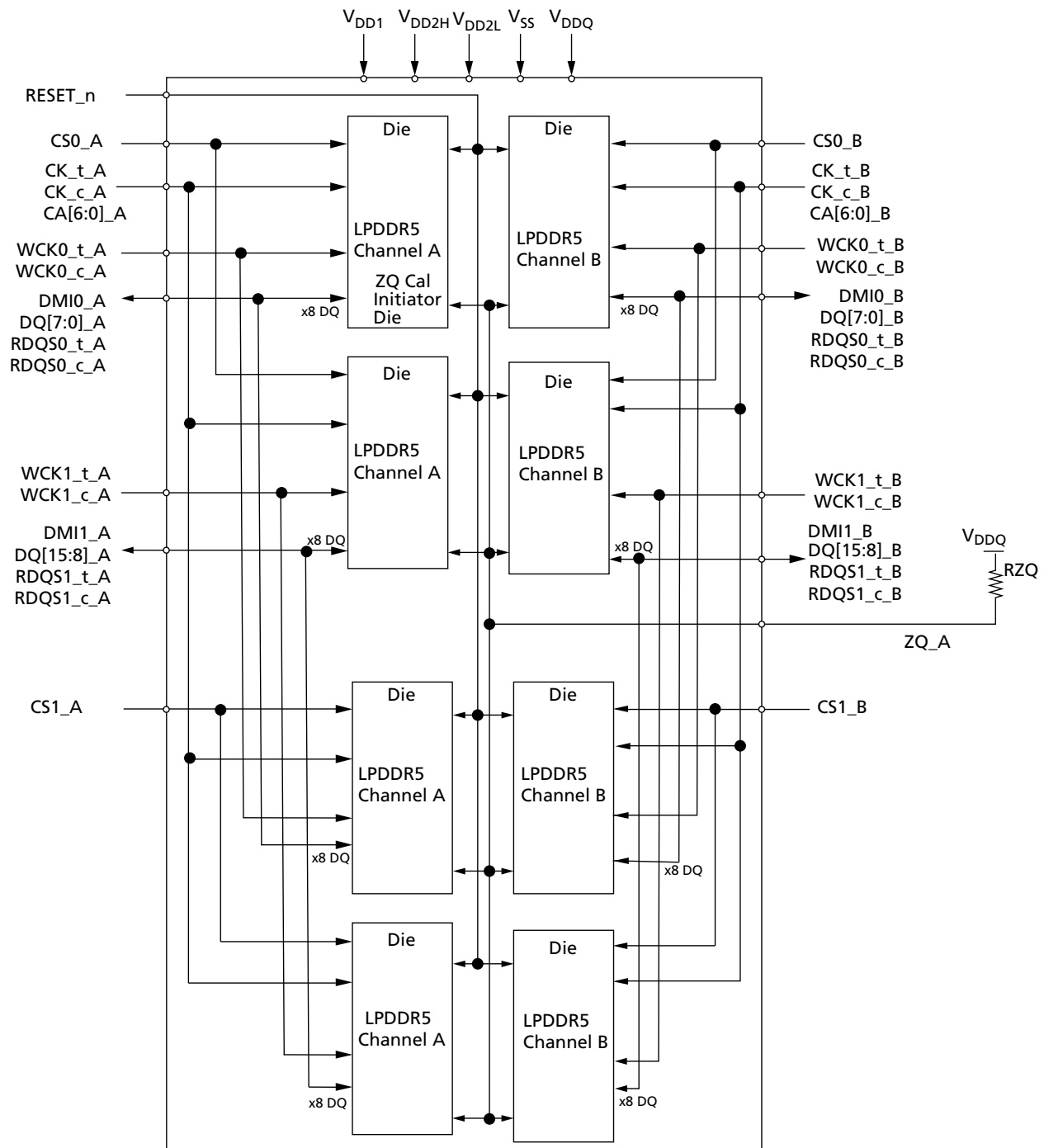
Quad Die, Dual Channel, Dual Rank

Figure 6: Quad-Die, Dual-Channel, Dual-Rank Package Block Diagram



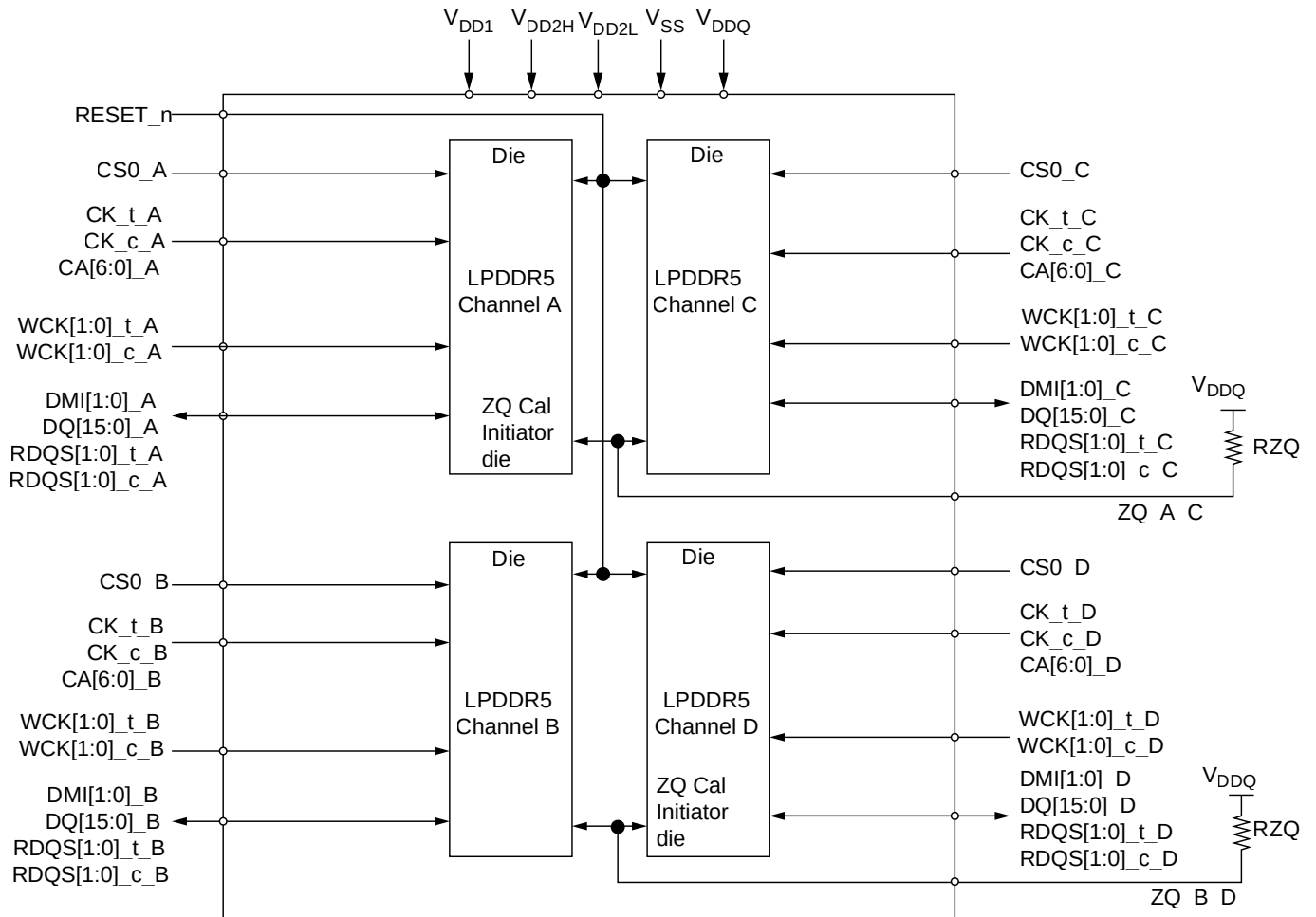
Eight Die, Dual Channel, Dual Rank

Figure 7: Eight-Die, Dual-Channel, Dual-Rank Package Block Diagram



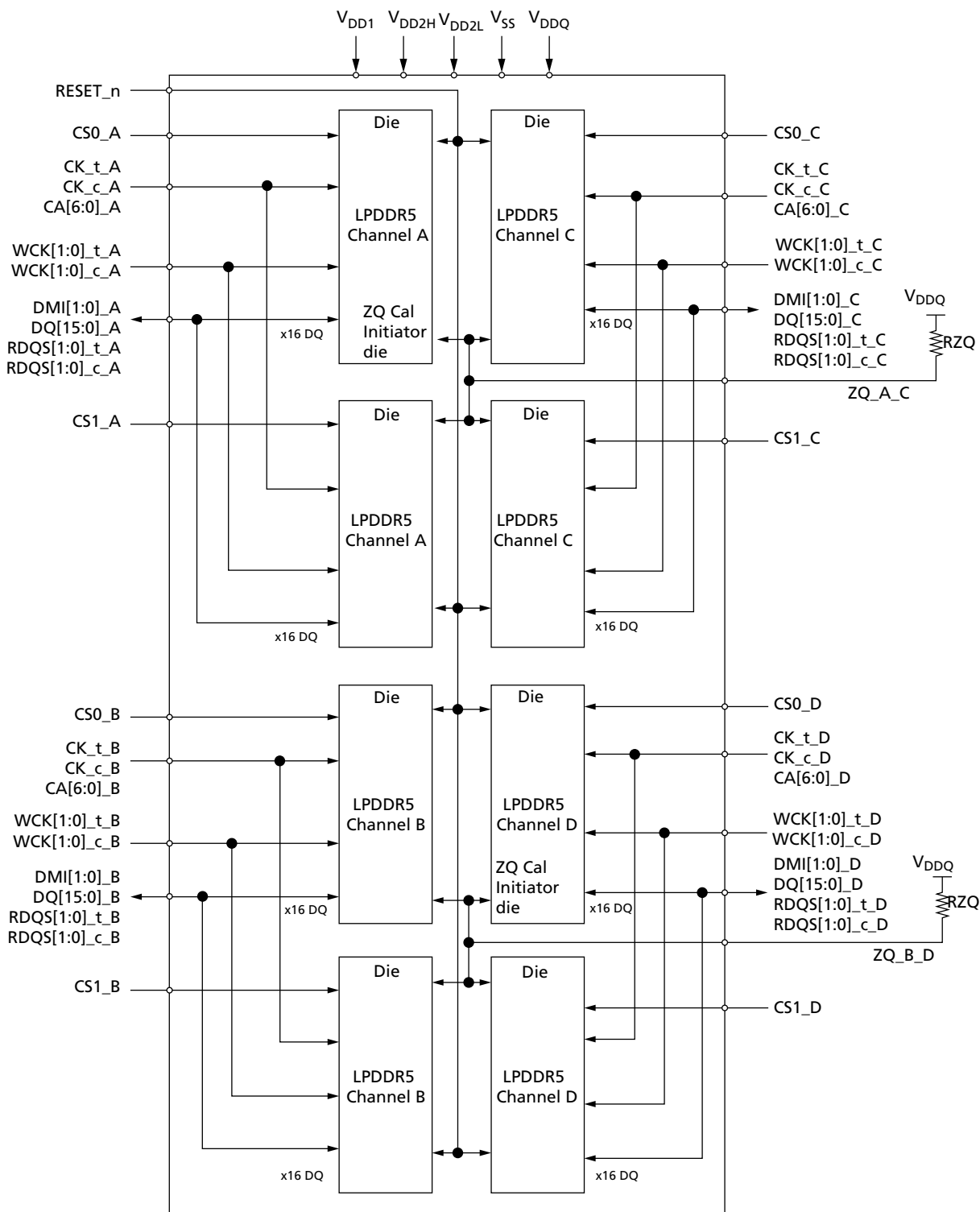
Quad Die, Quad Channel, Single Rank

Figure 8: Quad-Die, Quad-Channel, Single-Rank Package Block Diagram



Eight Die, Quad Channel, Dual Rank

Figure 9: Eight-Die, Quad-Channel, Dual-Rank Package Block Diagram



Ball Assignments and Descriptions

315b Single Die, Single Channel, 1 Rank, 2 Rank

Table 12: Single Channel 315-Ball/Pad Descriptions

Symbol	Type	Description
CK_t_[A], CK_c_[A]	Input	Clock: CK_t and CK_c are differential clock inputs. All double data rate (DDR) command/address inputs are sampled on both crossing points of CK_t and CK_c. The first crossing point is the rising (falling) edge of CK_t (CK_c) and second crossing point is falling (rising) edge of CK_t (CK_c). Single data rate (SDR) inputs, CS is sampled on the crossing point that is the rising (falling) edge of CK_t (CK_c).
CS0_[A], CS1_[A]	Input	Chip select: CS is part of the command code, and is sampled on the rising (falling) edge of CK_t (CK_c) unless the device is in power-down or deep sleep mode where it becomes an asynchronous signal. Each rank (0, 1) has its own CS signals. CS1_[A:B] become NC pins in a single-rank package.
CA[6:0]_[A]	Input	Command/address inputs: Provide the command and address inputs according to the command truth table.
WCK[1:0]_t_[A], WCK[1:0]_c_[A]	Input	Data clock: WCK_t and WCK_c are differential clock inputs used for write data capture and read data output.
DQ[15:0]_[A]	I/O	Data input/output: Bidirectional data bus.
RDQS[1:0]_t_[A], RDQS[1:0]_c_[A]	I/O Output	Read data strobe: RDQS_t and RDQS_c are differential output clock signals used to strobe data during a READ operation. RDQS_t is also used as a parity pin during write link protection enabled. Each byte of data has RDQS_t and RDQS_c signals.
DMI[1:0]_[A]	I/O	Data mask inversion: DMI serves multiple functions such as data mask (DM), data bus inversion (DBI), and parity at READ with ECC operation by setting the mode register. DMI is a bidirectional signal and each byte of data has a DMI signal.
ZQ_A	Reference	ZQ calibration reference: Used to calibrate the output drive strength and the termination resistance. The ZQ pin should be connected to V _{DDQ} through a 240 ohm $\pm 1\%$ resistor.
V _{DDQ} , V _{DD1} , V _{DD2H} , V _{DD2L}	Supply	Power supplies: Isolated on the die for improved noise immunity.
V _{SS}	Supply	Ground reference: Power supply ground reference.
RESET_n	Input	Reset: When asserted LOW, the RESET pin resets the die. Reset is an asynchronous signal.
NC	–	No connect: Not internally connected.
RFU	–	Reserved Future Use: Not internally connected

Figure 10: 315-Ball Single-Channel Discrete FBGA

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	
A	NC	NC	V _{DDQ}	DMI0_A	V _{SS}	V _{DD2L}	V _{DD2H}	V _{DD2H}	V _{DD2H}	V _{DD2L}	V _{SS}	DMI1_A	V _{DDQ}	NC	NC	A
B	NC	V _{DDQ}	RDQS0_t_A	V _{SS}	DQ4_A	V _{DD2L}	V _{DD2H}	V _{SS}	V _{DD2H}	V _{DD2L}	DQ12_A	V _{SS}	RDQS1_t_A	V _{DDQ}	NC	B
C	V _{DD1}	DQ1_A	V _{DDQ}	RDQS0_c_A	V _{SS}	DQ5_A	V _{DD2H}	V _{SS}	V _{DD2H}	DQ13_A	V _{SS}	RDQS1_c_A	V _{DDQ}	DQ9_A	V _{DD1}	C
D	DQ0_A	V _{SS}	DQ3_A	V _{DDQ}	WCK0_c_A	V _{SS}	V _{SS}	V _{DD2H}	V _{SS}	V _{SS}	WCK1_c_A	V _{DDQ}	DQ11_A	V _{SS}	DQ8_A	D
E	V _{SS}	DQ2_A	V _{SS}	WCK0_t_A	V _{DDQ}	DQ6_A	V _{DD2H}	V _{SS}	V _{DD2H}	DQ14_A	V _{DDQ}	WCK1_t_A	V _{SS}	DQ10_A	V _{SS}	E
F	V _{DDQ}	V _{SS}	V _{DDQ}	V _{DDQ}	DQ7_A	V _{DD2H}	V _{DD2H}	V _{SS}	V _{DD2H}	V _{DD2H}	DQ15_A	V _{DDQ}	V _{DDQ}	V _{SS}	V _{DDQ}	F
G	V _{DDQ}	V _{DDQ}	V _{SS}	CA0_A	V _{SS}	CS1_A	V _{SS}	CA2_A	V _{SS}	CA4_A	V _{SS}	CA6_A	V _{SS}	V _{DDQ}	V _{DDQ}	G
H	RESET_N	V _{DD2L}	V _{SS}	V _{SS}	CA1_A	V _{SS}	CS0_A	V _{SS}	CK_t_A	V _{SS}	CA3_A	V _{SS}	CA5_A	V _{DD2L}	ZQ_A	H
J	V _{SS}	V _{DD2L}	V _{SS}	RFU	V _{DD2H}	RFU	V _{SS}	V _{SS}	CK_c_A	V _{SS}	V _{DD2H}	V _{SS}	V _{SS}	V _{DD2L}	V _{SS}	J
K	V _{DD2H}	V _{DD2H}	V _{DD2H}	V _{DD2H}	V _{DD2H}	V _{DD2H}	V _{SS}	V _{SS}	V _{SS}	V _{DD2H}	V _{DD2H}	V _{DD2H}	V _{DD2H}	V _{DD2H}	V _{DD2H}	K
L	V _{SS}	V _{SS}	V _{SS}	V _{SS}	V _{SS}	V _{DD2H}	V _{DD2H}	V _{DD2H}	V _{DD2H}	V _{DD2H}	V _{SS}	V _{SS}	V _{SS}	V _{SS}	V _{SS}	L
M	V _{DD2H}	V _{DD2H}	V _{DD2H}	V _{DD2H}	V _{DD2H}	V _{DD2H}	V _{SS}	V _{SS}	V _{SS}	V _{DD2H}	V _{DD2H}	V _{DD2H}	V _{DD2H}	V _{DD2H}	V _{DD2H}	M
N	V _{SS}	V _{DD2L}	V _{SS}	V _{SS}	V _{DD2H}	V _{SS}	NC	V _{SS}	V _{SS}	V _{SS}	V _{DD2H}	V _{SS}	V _{SS}	V _{DD2L}	V _{SS}	N
P	RFU	V _{DD2L}	NC	V _{SS}	NC	V _{SS}	NC	V _{SS}	NC	V _{SS}	NC	V _{SS}	V _{SS}	V _{DD2L}	RFU	P
R	V _{DDQ}	V _{DDQ}	V _{SS}	NC	V _{SS}	NC	V _{SS}	NC	V _{SS}	NC	V _{SS}	NC	V _{SS}	V _{DDQ}	V _{DDQ}	R
T	V _{DDQ}	V _{SS}	V _{DDQ}	V _{DDQ}	NC	V _{DD2H}	V _{DD2H}	V _{SS}	V _{DD2H}	V _{DD2H}	NC	V _{DDQ}	V _{DDQ}	V _{SS}	V _{DDQ}	T
U	V _{SS}	NC	V _{SS}	NC	V _{DDQ}	NC	V _{DD2H}	V _{SS}	V _{DD2H}	NC	V _{DDQ}	NC	V _{SS}	NC	V _{SS}	U
V	NC	V _{SS}	NC	V _{DDQ}	NC	V _{SS}	V _{SS}	V _{DD2H}	V _{SS}	V _{SS}	NC	V _{DDQ}	NC	V _{SS}	NC	V
W	V _{DD1}	NC	V _{DDQ}	NC	V _{SS}	NC	V _{DD2H}	V _{SS}	V _{DD2H}	NC	V _{SS}	NC	V _{DDQ}	NC	V _{DD1}	W
Y	NC	V _{DDQ}	NC	V _{SS}	NC	V _{DD2L}	V _{DD2H}	V _{SS}	V _{DD2H}	V _{DD2L}	NC	V _{SS}	NC	V _{DDQ}	NC	Y
AA	NC	NC	V _{DDQ}	NC	V _{SS}	V _{DD2L}	V _{DD2H}	V _{DD2H}	V _{DD2H}	V _{DD2L}	V _{SS}	NC	V _{DDQ}	NC	NC	AA
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	

Top View (ball down)

V_{SS}
 V_{DD1}
 V_{DD2H}
 V_{DD2L}
 V_{DDQ}
 CK
 RDQS
 WCK
 DQ,DMI
 CA, CS, ZQ, RESET
 NC, RFU

315b Dual Channel, 1 Rank, 2 Rank

Table 13: 315-Ball/Pad Descriptions

Symbol	Type	Description
CK_t_[A:B], CK_c_[A:B]	Input	Clock: CK_t and CK_c are differential clock inputs. All double data rate (DDR) command/address inputs are sampled on both crossing points of CK_t and CK_c. The first crossing point is the rising (falling) edge of CK_t (CK_c) and second crossing point is falling (rising) edge of CK_t (CK_c). Single data rate (SDR) inputs, CS is sampled on the crossing point that is the rising (falling) edge of CK_t (CK_c).
CS0_[A:B], CS1_[A:B]	Input	Chip select: CS is part of the command code, and is sampled on the rising (falling) edge of CK_t (CK_c) unless the device is in power-down or deep sleep mode where it becomes an asynchronous signal. Each rank (0, 1) has its own CS signals. CS1_[A:B] become NC pins in a single-rank package.
CA[6:0]_[A:B]	Input	Command/address inputs: Provide the command and address inputs according to the command truth table.
WCK[1:0]_t_[A:B], WCK[1:0]_c_[A:B]	Input	Data clock: WCK_t and WCK_c are differential clock inputs used for write data capture and read data output.
DQ[15:0]_[A:B]	I/O	Data input/output: Bidirectional data bus.
RDQS[1:0]_t_[A:B], RDQS[1:0]_c_[A:B]	I/O Output	Read data strobe: RDQS_t and RDQS_c are differential output clock signals used to strobe data during a READ operation. RDQS_t is also used as a parity pin during write link protection enabled. Each byte of data has RDQS_t and RDQS_c signals.
DMI[1:0]_[A:B]	I/O	Data mask inversion: DMI serves multiple functions such as data mask (DM), data bus inversion (DBI), and parity at READ with ECC operation by setting the mode register. DMI is a bidirectional signal and each byte of data has a DMI signal.
ZQ_A	Reference	ZQ calibration reference: Used to calibrate the output drive strength and the termination resistance. The ZQ pin should be connected to V _{DDQ} through a 240 ohm $\pm 1\%$ resistor.
V _{DDQ} , V _{DD1} , V _{DD2H} , V _{DD2L}	Supply	Power supplies: Isolated on the die for improved noise immunity.
V _{SS}	Supply	Ground reference: Power supply ground reference.
RESET_n	Input	Reset: When asserted LOW, the RESET pin resets the die. Reset is an asynchronous signal.
NC	–	No connect: Not internally connected.
RFU	–	Reserved Future Use: Not internally connected.

Figure 11: 315-Ball Dual-Channel Discrete FBGA

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	
A	NC	NC	V _{DDQ}	DMIO_A	V _{SS}	V _{DD2L}	V _{DD2H}	V _{DD2H}	V _{DD2H}	V _{DD2L}	V _{SS}	DMIO_A	V _{DDQ}	NC	NC	A
B	NC	V _{DDQ}	RDQS0_t_A	V _{SS}	DQ4_A	V _{DD2L}	V _{DD2H}	V _{SS}	V _{DD2H}	V _{DD2L}	DQ12_A	V _{SS}	RDQS1_t_A	V _{DDQ}	NC	B
C	V _{DD1}	DQ1_A	V _{DDQ}	RDQS0_c_A	V _{SS}	DQ5_A	V _{DD2H}	V _{SS}	V _{DD2H}	DQ13_A	V _{SS}	RDQS1_c_A	V _{DDQ}	DQ9_A	V _{DD1}	C
D	DQ0_A	V _{SS}	DQ3_A	V _{DDQ}	WCK0_c_A	V _{SS}	V _{SS}	V _{DD2H}	V _{SS}	V _{SS}	WCK1_c_A	V _{DDQ}	DQ11_A	V _{SS}	DQ8_A	D
E	V _{SS}	DQ2_A	V _{SS}	WCK0_t_A	V _{DDQ}	DQ6_A	V _{DD2H}	V _{SS}	V _{DD2H}	DQ14_A	V _{DDQ}	WCK1_t_A	V _{SS}	DQ10_A	V _{SS}	E
F	V _{DDQ}	V _{SS}	V _{DDQ}	V _{DDQ}	DQ7_A	V _{DD2H}	V _{DD2H}	V _{SS}	V _{DD2H}	V _{DD2H}	DQ15_A	V _{DDQ}	V _{DDQ}	V _{SS}	V _{DDQ}	F
G	V _{DDQ}	V _{DDQ}	V _{SS}	CA0_A	V _{SS}	CS1_A	V _{SS}	CA2_A	V _{SS}	CA4_A	V _{SS}	CA6_A	V _{SS}	V _{DDQ}	V _{DDQ}	G
H	RESET_N	V _{DD2L}	V _{SS}	V _{SS}	CA1_A	V _{SS}	CS0_A	V _{SS}	CK_t_A	V _{SS}	CA3_A	V _{SS}	CA5_A	V _{DD2L}	ZQ_A	H
J	V _{SS}	V _{DD2L}	V _{SS}	RFU	V _{DD2H}	RFU	V _{SS}	V _{SS}	CK_c_A	V _{SS}	V _{DD2H}	V _{SS}	V _{SS}	V _{DD2L}	V _{SS}	J
K	V _{DD2H}	V _{DD2H}	V _{DD2H}	V _{DD2H}	V _{DD2H}	V _{DD2H}	V _{SS}	V _{SS}	V _{SS}	V _{DD2H}	V _{DD2H}	V _{DD2H}	V _{DD2H}	V _{DD2H}	V _{DD2H}	K
L	V _{SS}	V _{SS}	V _{SS}	V _{SS}	V _{SS}	V _{DD2H}	V _{DD2H}	V _{DD2H}	V _{DD2H}	V _{DD2H}	V _{SS}	V _{SS}	V _{SS}	V _{SS}	V _{SS}	L
M	V _{DD2H}	V _{DD2H}	V _{DD2H}	V _{DD2H}	V _{DD2H}	V _{DD2H}	V _{SS}	V _{SS}	V _{SS}	V _{DD2H}	V _{DD2H}	V _{DD2H}	V _{DD2H}	V _{DD2H}	V _{DD2H}	M
N	V _{SS}	V _{DD2L}	V _{SS}	V _{SS}	V _{DD2H}	V _{SS}	CK_c_B	V _{SS}	V _{SS}	V _{SS}	V _{DD2H}	V _{SS}	V _{SS}	V _{DD2L}	V _{SS}	N
P	RFU	V _{DD2L}	CA5_B	V _{SS}	CA3_B	V _{SS}	CK_t_B	V _{SS}	CS0_B	V _{SS}	CA1_B	V _{SS}	V _{SS}	V _{DD2L}	RFU	P
R	V _{DDQ}	V _{DDQ}	V _{SS}	CA6_B	V _{SS}	CA4_B	V _{SS}	CA2_B	V _{SS}	CS1_B	V _{SS}	CA0_B	V _{SS}	V _{DDQ}	V _{DDQ}	R
T	V _{DDQ}	V _{SS}	V _{DDQ}	V _{DDQ}	DQ15_B	V _{DD2H}	V _{DD2H}	V _{SS}	V _{DD2H}	V _{DD2H}	DQ7_B	V _{DDQ}	V _{DDQ}	V _{SS}	V _{DDQ}	T
U	V _{SS}	DQ10_B	V _{SS}	WCK1_t_B	V _{DDQ}	DQ14_B	V _{DD2H}	V _{SS}	V _{DD2H}	DQ6_B	V _{DDQ}	WCK0_t_B	V _{SS}	DQ2_B	V _{SS}	U
V	DQ8_B	V _{SS}	DQ11_B	V _{DDQ}	WCK1_c_B	V _{SS}	V _{SS}	V _{DD2H}	V _{SS}	V _{SS}	WCK0_c_B	V _{DDQ}	DQ3_B	V _{SS}	DQ0_B	V
W	V _{DD1}	DQ9_B	V _{DDQ}	RDQS1_t_B	V _{SS}	DQ13_B	V _{DD2H}	V _{SS}	V _{DD2H}	DQ5_B	V _{SS}	RDQS0_c_B	V _{DDQ}	DQ1_B	V _{DD1}	W
Y	NC	V _{DDQ}	RDQS1_t_B	V _{SS}	DQ12_B	V _{DD2L}	V _{DD2H}	V _{SS}	V _{DD2H}	V _{DD2L}	DQ4_B	V _{SS}	RDQS0_t_B	V _{DDQ}	NC	Y
AA	NC	NC	V _{DDQ}	DMIO_B	V _{SS}	V _{DD2L}	V _{DD2H}	V _{DD2H}	V _{DD2H}	V _{DD2L}	V _{SS}	DMIO_B	V _{DDQ}	NC	NC	AA
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	

Top View (ball down)

V_{SS}
V_{DD1}
V_{DD2H}
V_{DD2L}
V_{DDQ}
CK
RDQS
WCK
DQ,DMI
CA, CS, ZQ, RESET
NC, RFU

441-Ball Quad Channel, 1 Rank, 2 Rank

Table 14: 441-Ball/Pad Descriptions

Symbol	Type	Description
CK_t_[A:D], CK_c_[A:D]	Input	Clock: CK_t and CK_c are differential clock inputs. All double data rate (DDR) command/address inputs are sampled on both crossing points of CK_t and CK_c. The first crossing point is the rising (falling) edge of CK_t (CK_c) and second crossing point is falling (rising) edge of CK_t (CK_c). Single data rate (SDR) inputs, CS is sampled on the crossing point that is the rising (falling) edge of CK_t (CK_c).
CS0_[A:D], CS1_[A:D]	Input	Chip select: CS is part of the command code, and is sampled on the rising (falling) edge of CK_t (CK_c) unless the device is in power-down or deep sleep mode where it becomes an asynchronous signal. Each rank (0, 1) has its own CS signals. CS1_[A:D] become NC pins in a single-rank package.
CA[6:0]_[A:D]	Input	Command/address inputs: Provide the command and address inputs according to the command truth table.
WCK[1:0]_t_[A:D], WCK[1:0]_c_[A:D]	Input	Data clock: WCK_t and WCK_c are differential clock inputs used for WRITE data capture and READ data output.
DQ[15:0]_[A:D]	I/O	Data input/output: Bidirectional data bus.
RDQS[1:0]_t_[A:D], RDQS[1:0]_c_[A:D]	I/O Output	Read data strobe: RDQS_t and RDQS_c are differential output clock signals used to strobe data during a READ operation. RDQS_t is also used as a parity pin during write link protection enabled. Each byte of data has RDQS_t and RDQS_c signals.
DMI[1:0]_[A:D]	I/O	Data mask inversion: DMI serves multiple functions such as data mask (DM), data bus inversion (DBI), and parity at READ with ECC operation by setting the mode register. DMI is a bidirectional signal and each byte of data has a DMI signal.
ZQ_A_C, ZQ_B_D	Reference	ZQ calibration reference: Used to calibrate the output drive strength and the termination resistance. The ZQ pin should be connected to V _{DDQ} through a 240 ohm $\pm 1\%$ resistor.
V _{DDQ} , V _{DD1} , V _{DD2H} , V _{DD2L}	Supply	Power supplies: Isolated on the die for improved noise immunity.
V _{SS}	Supply	Ground reference: Power supply ground reference.
RESET_n	Input	Reset: When asserted LOW, the RESET pin resets the die. Reset is an asynchronous signal.
NC	–	No connect: Not internally connected.
RFU	–	Reserved Future Use: Not internally connected

Figure 12: 441-Ball Quad-Channel FBGA

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	
A	V _{SS}	V _{SS}	V _{DD1}	V _{DD2L}	V _{SS}	V _{DD2H}	V _{DD1}	V _{SS}	V _{DD2L}	V _{DD2H}	V _{DD2H}	V _{SS}	V _{DD1}	V _{DD2L}	V _{SS}	V _{DD2H}	V _{DD1}	V _{SS}	V _{DD2L}	V _{SS}	V _{SS}	A
B	V _{SS}	DQ0_A	V _{SS}	DQ3_A	V _{DD2H}	V _{SS}	DQ11_A	DQ9_A	DQ8_A	V _{SS}	V _{DD2H}	DQ0_C	V _{SS}	DQ3_C	V _{DD2H}	V _{SS}	DQ11_C	DQ9_C	DQ8_C	RFU	V _{SS}	B
C	V _{DD2H}	V _{SS}	DQ2_A	V _{DDQ}	CA0_A	V _{DD2H}	V _{SS}	DQ10_A	V _{DDQ}	V _{DD2H}	V _{SS}	V _{SS}	DQ2_C	V _{DDQ}	CA0_C	V _{DD2H}	V _{SS}	DQ10_C	V _{DDQ}	V _{DD2H}	V _{DD2H}	C
D	V _{SS}	DQ1_A	WCK0_c_A	V _{SS}	CA1_A	CS0_A	V _{DDQ}	V _{SS}	WCK1_t_A	V _{DD2H}	V _{DDQ}	DQ1_C	WCK0_c_C	V _{SS}	CA1_C	CS0_C	V _{DDQ}	V _{SS}	WCK1_t_C	V _{DDQ}	V _{SS}	D
E	V _{DDQ}	RDQS0_c_A	V _{SS}	WCK0_t_A	V _{SS}	CS1_A	V _{SS}	WCK1_c_A	DMI1_A	V _{SS}	V _{DDQ}	RDQS0_c_C	V _{SS}	WCK0_t_C	V _{SS}	CS1_C	V _{SS}	WCK1_c_C	DMI1_C	V _{SS}	V _{DD2H}	E
F	V _{DDQ}	RDQS0_t_A	V _{SS}	V _{DDQ}	V _{SS}	CA2_A	V _{SS}	RDQS1_t_A	V _{SS}	V _{DDQ}	V _{SS}	RDQS0_t_C	V _{SS}	V _{DDQ}	V _{SS}	CA2_C	V _{SS}	RDQS1_t_C	V _{SS}	V _{DDQ}	V _{DD2H}	F
G	V _{SS}	DQ4_A	V _{DDQ}	DMI0_A	RFU	RFU	CA6_A	V _{SS}	RDQS1_c_A	V _{SS}	V _{DDQ}	DMI0_C	V _{DDQ}	DQ4_C	RFU	RFU	CA6_C	V _{SS}	RDQS1_c_C	V _{SS}	V _{SS}	G
H	V _{DD2L}	V _{SS}	DQ5_A	V _{SS}	CK_t_A	V _{SS}	CA5_A	V _{DDQ}	V _{SS}	DQ12_A	V _{SS}	V _{SS}	DQ5_C	V _{SS}	CK_t_C	V _{SS}	CA5_C	V _{DDQ}	V _{SS}	DQ12_C	V _{DD2L}	H
J	V _{DD2H}	DQ6_A	DQ7_A	V _{DD2H}	V _{SS}	CK_c_A	V _{SS}	DQ14_A	DQ13_A	V _{SS}	V _{DD2L}	DQ6_C	DQ7_C	V _{DD2L}	ZQ_A_C	CK_c_C	V _{SS}	DQ14_C	DQ13_C	V _{SS}	V _{DD2H}	J
K	V _{SS}	V _{DD2H}	V _{DD2H}	V _{DD2H}	V _{SS}	CA3_A	CA4_A	V _{DD2L}	V _{SS}	DQ15_A	V _{DD2H}	V _{DD2H}	V _{DD2H}	V _{DD2H}	V _{SS}	CA3_C	CA4_C	V _{DD2H}	V _{SS}	DQ15_C	V _{SS}	K
L	V _{DD2H}	V _{DD2L}	V _{DD2L}	V _{DD2H}	V _{DD2L}	V _{DD2H}	V _{SS}	V _{DD2H}	V _{DD2H}	V _{SS}	V _{SS}	V _{SS}	V _{DD2H}	V _{DD2H}	V _{SS}	V _{DD2H}	V _{DD2L}	V _{DD2H}	V _{DD2L}	V _{DD2L}	V _{DD2H}	L
M	V _{SS}	DQ15_B	V _{SS}	V _{DD2H}	CA4_B	CA3_B	V _{SS}	V _{DD2H}	V _{DD2H}	V _{DD2H}	V _{DD2H}	DQ15_D	V _{SS}	V _{DD2L}	CA4_D	CA3_D	V _{SS}	V _{DD2H}	V _{DD2H}	V _{DD2H}	V _{SS}	M
N	V _{DD2H}	V _{SS}	DQ13_B	DQ14_B	V _{SS}	CK_c_B	ZQ_B_D	V _{DD2L}	DQ7_B	DQ6_B	V _{DD2L}	V _{SS}	DQ13_D	DQ14_D	V _{SS}	CK_c_D	V _{SS}	V _{DD2H}	DQ7_D	DQ6_D	V _{DD2H}	N
P	V _{DD2L}	DQ12_B	V _{SS}	V _{DDQ}	CA5_B	V _{SS}	CK_t_B	V _{SS}	DQ5_B	V _{SS}	V _{SS}	DQ12_D	V _{SS}	V _{DDQ}	CA5_D	V _{SS}	CK_t_D	V _{SS}	DQ5_D	V _{SS}	V _{DD2L}	P
R	V _{SS}	V _{SS}	RDQS1_c_B	V _{SS}	CA6_B	RFU	RFU	DQ4_B	V _{DDQ}	DMI0_B	V _{DDQ}	V _{SS}	RDQS1_c_D	V _{SS}	CA6_D	RFU	RFU	DMI0_D	V _{DDQ}	DQ4_D	V _{SS}	R
T	V _{DD2H}	V _{DDQ}	V _{SS}	RDQS1_t_B	V _{SS}	CA2_B	V _{SS}	V _{DDQ}	V _{SS}	RDQS0_t_B	V _{SS}	V _{DDQ}	V _{SS}	RDQS1_t_D	V _{SS}	CA2_D	V _{SS}	V _{DDQ}	V _{SS}	RDQS0_t_D	V _{DDQ}	T
U	V _{DD2H}	V _{SS}	DMI1_B	WCK1_c_B	V _{SS}	CS1_B	V _{SS}	WCK0_t_B	V _{SS}	RDQS0_c_B	V _{DDQ}	V _{SS}	DMI1_D	WCK1_c_D	V _{SS}	CS1_D	V _{SS}	WCK0_t_D	V _{SS}	RDQS0_c_D	V _{DDQ}	U
V	V _{SS}	V _{DDQ}	WCK1_t_B	V _{SS}	V _{DDQ}	CS0_B	CA1_B	V _{SS}	WCK0_c_B	DQ1_B	V _{DDQ}	V _{DD2H}	WCK1_t_D	V _{SS}	V _{DDQ}	CS0_D	CA1_D	V _{SS}	WCK0_c_D	DQ1_D	V _{SS}	V
W	V _{DD2H}	V _{DD2H}	V _{DDQ}	DQ10_B	V _{SS}	V _{DD2H}	CA0_B	V _{DDQ}	DQ2_B	V _{SS}	V _{SS}	V _{DD2H}	V _{DDQ}	DQ10_D	V _{SS}	V _{DD2H}	CA0_D	V _{DDQ}	DQ2_D	V _{SS}	V _{DD2H}	W
Y	V _{SS}	RESET_N	DQ8_B	DQ9_B	DQ11_B	V _{SS}	V _{DD2H}	DQ3_B	V _{SS}	DQ0_B	V _{DD2H}	V _{SS}	DQ8_D	DQ9_D	DQ11_D	V _{SS}	V _{DD2H}	DQ3_D	V _{SS}	DQ0_D	V _{SS}	Y
AA	V _{SS}	V _{SS}	V _{DD2L}	V _{SS}	V _{DD1}	V _{DD2H}	V _{SS}	V _{DD2L}	V _{DD1}	V _{SS}	V _{DD2H}	V _{DD2H}	V _{DD2L}	V _{SS}	V _{DD1}	V _{DD2H}	V _{SS}	V _{DD2L}	V _{DD1}	V _{SS}	V _{SS}	AA

Top View (ball down)

 V _{SS}	 V _{DD1}	 V _{DD2H}	 V _{DD2L}	 V _{DDQ}	 CK	 RDQS	 WCK	 DQ, DMI	 CA, CS, ZQ, RESET	 NC
---	--	---	--	--	--	--	---	---	---	--

561-Ball Quad Channel, 1 Rank, 2 Rank

Table 15: Ball/Pad Descriptions

Symbol	Type	Description
CK_t_[A:D] CK_c_[A:D]	Input	Clock: CK_t and CK_c are differential clock inputs. All double data rate (DDR) command/address inputs are sampled on both crossing points of CK_t and CK_c. The first crossing point is the rising (falling) edge of CK_t (CK_c) and second crossing point is falling (rising) edge of CK_t (CK_c). Single data rate (SDR) inputs, CS is sampled on the crossing point that is the rising (falling) edge of CK_t (CK_c).
CS0_[A:D], CS1_[A:D]	Input	Chip select: CS is part of the command code, and is sampled on the rising (falling) edge of CK_t (CK_c) unless the device is in power-down or deep sleep mode where it becomes an asynchronous signal. Each rank (0, 1) has its own CS signals. CS1_[A:D] become NC pins in a single-rank package.
CA[6:0]_[A:D]	Input	Command/address inputs: Provide the command and address inputs according to the command truth table.
WCK[1:0]_t_[A:D] WCK[1:0]_c_[A:D]	Input	Data clock: WCK_t and WCK_c are differential clock inputs used for WRITE data capture and READ data output.
DQ[15:0]_[A:D]	I/O	Data input/output: Bidirectional data bus.
RDQS[1:0]_t_[A:D] RDQS[1:0]_c_[A:D]	I/O Output	Read data strobe: RDQS_t and RDQS_c are differential output clock signals used to strobe data during a READ operation. RDQS_t is also used as a parity pin during write link protection enabled. Each byte of data has RDQS_t and RDQS_c signals.
DMI[1:0]_[A:D]	I/O	Data mask inversion: DMI serves multiple functions such as data mask (DM), data bus inversion (DBI), and parity at READ with ECC operation by setting the mode register. DMI is a bidirectional signal and each byte of data has a DMI signal.
ZQ_A_C, ZQ_B_D	Reference	ZQ calibration reference: Used to calibrate the output drive strength and the termination resistance. The ZQ pin should be connected to V _{DDQ} through a 240 ohm $\pm 1\%$ resistor.
V _{DDQ} , V _{DD1} , V _{DD2H} , V _{DD2L}	Supply	Power supplies: Isolated on the die for improved noise immunity.
V _{SS}	Supply	Ground reference: Power supply ground reference.
RESET_n	Input	Reset: When asserted LOW, the RESET pin resets the die. Reset is an asynchronous signal.
NC	–	No connect: Not internally connected.

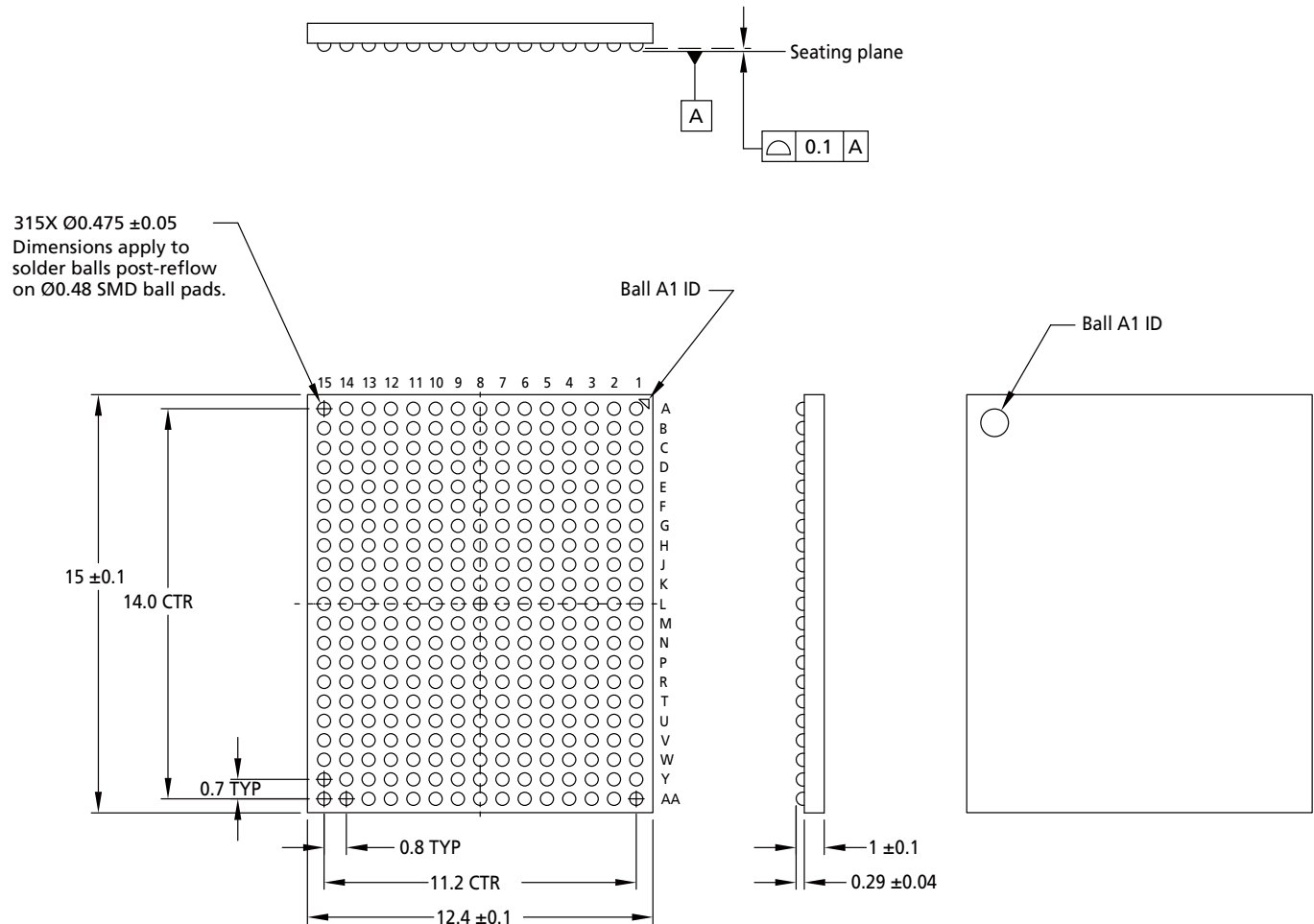
Figure 13: 561-Ball Quad-Channel FBGA

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	
A		NC	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	NC	NC	A
B	NC	VSS	VSS	VSS	VSS	VSS	VSS	VDD2H	VDD2H	VSS	VDD2H	VDD2H	VSS	VSS	VSS	VSS	VSS	VSS	NC	B
C	VSS	VSS	DQ0_A	VDDQ	VSS	DMI0_A	DQ5_A	VSS	CS0_A	CSL_A	CA3_A	VSS	DQ13_A	DMI1_A	VSS	VDDQ	DQ8_A	VSS	VSS	C
D	VSS	VDD1	VSS	DQ3_A	WCK0_C_A	VDDQ	VSS	CA0_A	VSS	VSS	VSS	CA6_A	VSS	VDDQ	WCK1_C_A	DQ11_A	VSS	VDD1	VSS	D
E	VSS	VDD2H	DQ1_A	VSS	WCK0_T_A	VSS	DQ6_A	VDDQ	CA1_A	CK_T_A	CA4_A	VDDQ	DQ14_A	VSS	WCK1_T_A	VSS	DQ9_A	VDD2H	VSS	E
F	VSS	VSS	VDDQ	RDQS0_T_A	VDD2H	DQ4_A	VDDQ	VSS	VSS	CK_C_A	VSS	VSS	VDDQ	DQ12_A	VDD2H	RDQS1_T_A	VDDQ	VSS	VSS	F
G	VSS	VDD2L	DQ2_A	RDQS0_C_A	VSS	VDD2H	DQ7_A	VDD2H	CA2_A	VDD2L	CA5_A	VDD2H	DQ15_A	VDD2H	VSS	RDQS1_C_A	DQ10_A	ZQ_A_C	VSS	G
H	VSS	VDD2H	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VDD2L	VSS	H
J	VSS	VDD2L	DQ2_C	RDQS0_C_C	VSS	VDD2H	DQ7_C	VDD2H	CA2_C	VDD2L	CA5_C	VDD2H	DQ15_C	VDD2H	VSS	RDQS1_C_C	DQ10_C	RESET_N	VSS	J
K	VSS	VSS	VDDQ	RDQS0_T_C	VDD2H	DQ4_C	VDDQ	VSS	VSS	CK_C_C	VSS	VSS	VDDQ	DQ12_C	VDD2H	RDQS1_T_C	VDDQ	VSS	VSS	K
L	VSS	VDD2H	DQ1_C	VSS	WCK0_T_C	VSS	DQ6_C	VDDQ	CA1_C	CK_T_C	CA4_C	VDDQ	DQ14_C	VSS	WCK1_T_C	VSS	DQ9_C	VDD2H	VSS	L
M	VSS	VSS	VSS	DQ3_C	WCK0_C_C	VDDQ	VSS	CA0_C	VSS	VSS	VSS	CA6_C	VSS	VDDQ	WCK1_C_C	DQ11_C	VSS	VSS	VSS	M
N	VSS	VDD1	DQ0_C	VDDQ	VSS	DMI0_C	DQ5_C	VSS	CS0_C	CSL_C	CA3_C	VSS	DQ13_C	DMI1_C	VSS	VDDQ	DQ8_C	VDD1	VSS	N
P	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VDD2H	VDD2H	VSS	VDD2H	VDD2H	VSS	VSS	VSS	VSS	VSS	VSS	VSS	P
R	VSS	VSS					VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	R
T	VSS	VSS					VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	T
U	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VDD2H	VDD2H	VSS	VDD2H	VDD2H	VSS	VSS	VSS	VSS	VSS	VSS	VSS	U
V	VSS	VDD1	DQ8_D	VDDQ	VSS	DMI1_D	DQ13_D	VSS	CA3_D	CSL_D	CS0_D	VSS	DQ5_D	DMI0_D	VSS	VDDQ	DQ0_D	VDD1	VSS	V
W	VSS	VSS	VSS	DQ11_D	WCK1_C_D	VDDQ	VSS	CA6_D	VSS	VSS	VSS	CA0_D	VSS	VDDQ	WCK0_C_D	DQ3_D	VSS	VSS	VSS	W
Y	VSS	VDD2H	DQ9_D	VSS	WCK1_T_D	VSS	DQ14_D	VDDQ	CA4_D	CK_T_D	CA1_D	VDDQ	DQ6_D	VSS	WCK0_T_D	VSS	DQ1_D	VDD2H	VSS	Y
AA	VSS	VSS	VDDQ	RDQS1_T_D	VDD2H	DQ12_D	VDDQ	VSS	VSS	CK_C_D	VSS	VSS	VDDQ	DQ4_D	VDD2H	RDQS0_T_D	VDDQ	VSS	VSS	AA
AB	VSS	VDD2L	DQ10_D	RDQS1_C_D	VSS	VDD2H	DQ15_D	VDD2H	CA5_D	VDD2L	CA2_D	VDD2H	DQ7_D	VDD2H	VSS	RDQS0_C_D	DQ2_D	VSS	VSS	AB
AC	VSS	VDD2H	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VDD2L	VSS	AC
AD	VSS	VDD2L	DQ10_B	RDQS1_C_B	VSS	VDD2H	DQ15_B	VDD2H	CA5_B	VDD2L	CA2_B	VDD2H	DQ7_B	VDD2H	VSS	RDQS0_C_B	DQ2_B	ZQ_B_D	VSS	AD
AE	VSS	VSS	VDDQ	RDQS1_T_B	VDD2H	DQ12_B	VDDQ	VSS	VSS	CK_C_B	VSS	VSS	VDDQ	DQ4_B	VDD2H	RDQS0_T_B	VDDQ	VSS	VSS	AE
AF	VSS	VDD2H	DQ9_B	VSS	WCK1_T_B	VSS	DQ14_B	VDDQ	CA4_B	CK_T_B	CA1_B	VDDQ	DQ6_B	VSS	WCK0_T_B	VSS	DQ1_B	VDD2H	VSS	AF
AG	VSS	VDD1	VSS	DQ11_B	WCK1_C_B	VDDQ	VSS	CA6_B	VSS	VSS	VSS	CA0_B	VSS	VDDQ	WCK0_C_B	DQ3_B	VSS	VDD1	VSS	AG
AH	VSS	VSS	DQ8_B	VDDQ	VSS	DMI1_B	DQ13_B	VSS	CA3_B	CSL_B	CS0_B	VSS	DQ5_B	DMI0_B	VSS	VDDQ	DQ0_B	VSS	VSS	AH
AJ	NC	VSS	VSS	VSS	VSS	VSS	VSS	VDD2H	VDD2H	VSS	VDD2H	VDD2H	VSS	VSS	VSS	VSS	VSS	VSS	NC	AJ
AK	NC	NC	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	VSS	NC	NC	AK
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	

Package Dimensions

315-Ball Package (Package Code: DS)

Figure 14: 315-Ball TFBGA – 12.4mm (TYP) × 15.0mm (TYP) × 1.1mm (MAX)

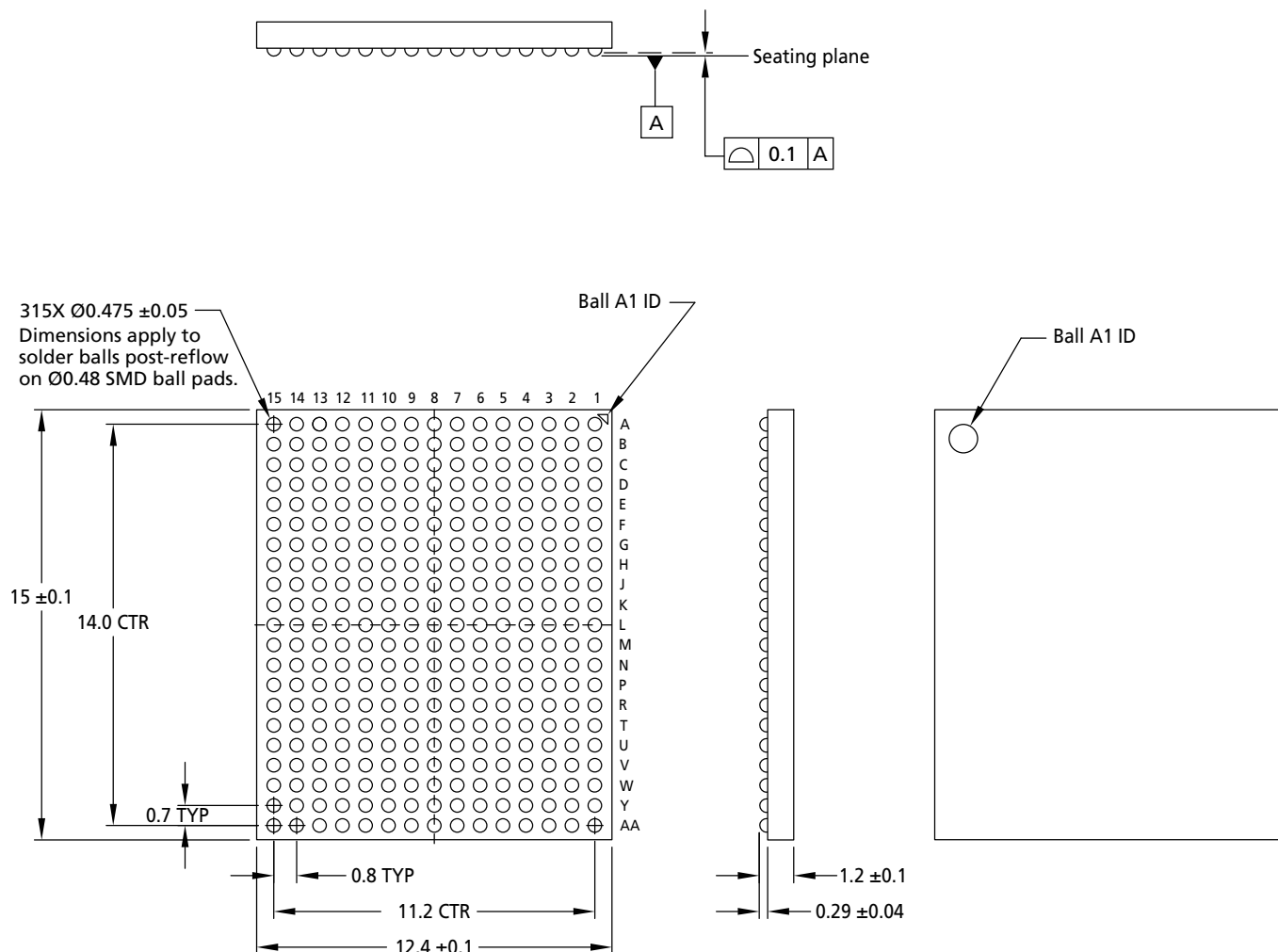


Notes: 1. All dimensions are in millimeters.

2. Solder ball composition: SACQ with CuOSP pads (Sn-4Ag-0.5Cu-3Bi-0.05Ni)

315-Ball Package (Package Code: DV)

Figure 15: 315-Ball LFBGA – 12.4mm (TYP) × 15.0mm (TYP) × 1.3mm (MAX)

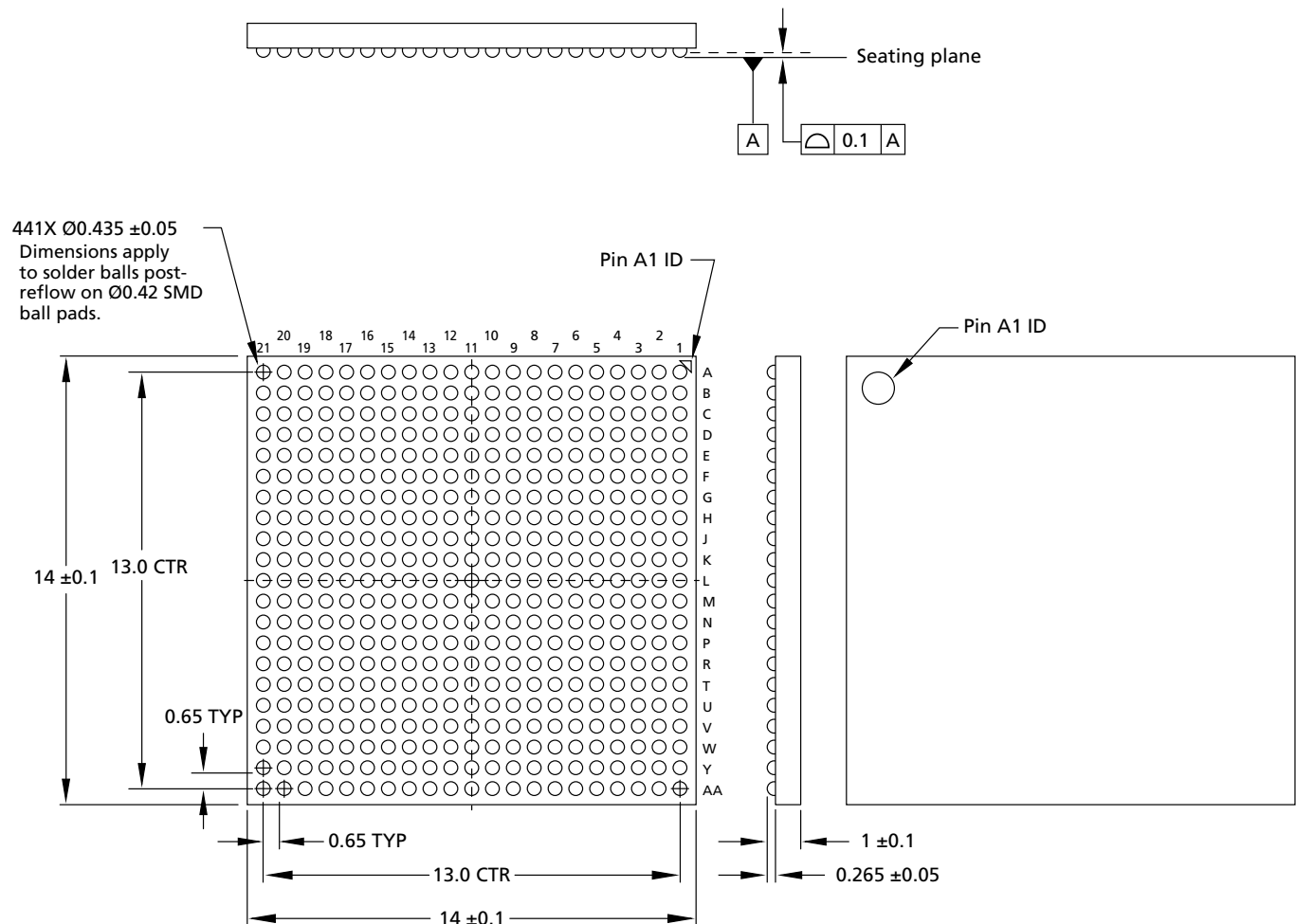


Notes: 1. All dimensions are in millimeters.

2. Solder ball composition: SACQ with CuOSP pads (Sn-4Ag-0.5Cu-3Bi-0.05Ni)

441-Ball Package (Package Code: EK)

Figure 16: 441-Ball TFBGA – 14.0mm (TYP) x 14.0mm (TYP) x 1.1mm (MAX)

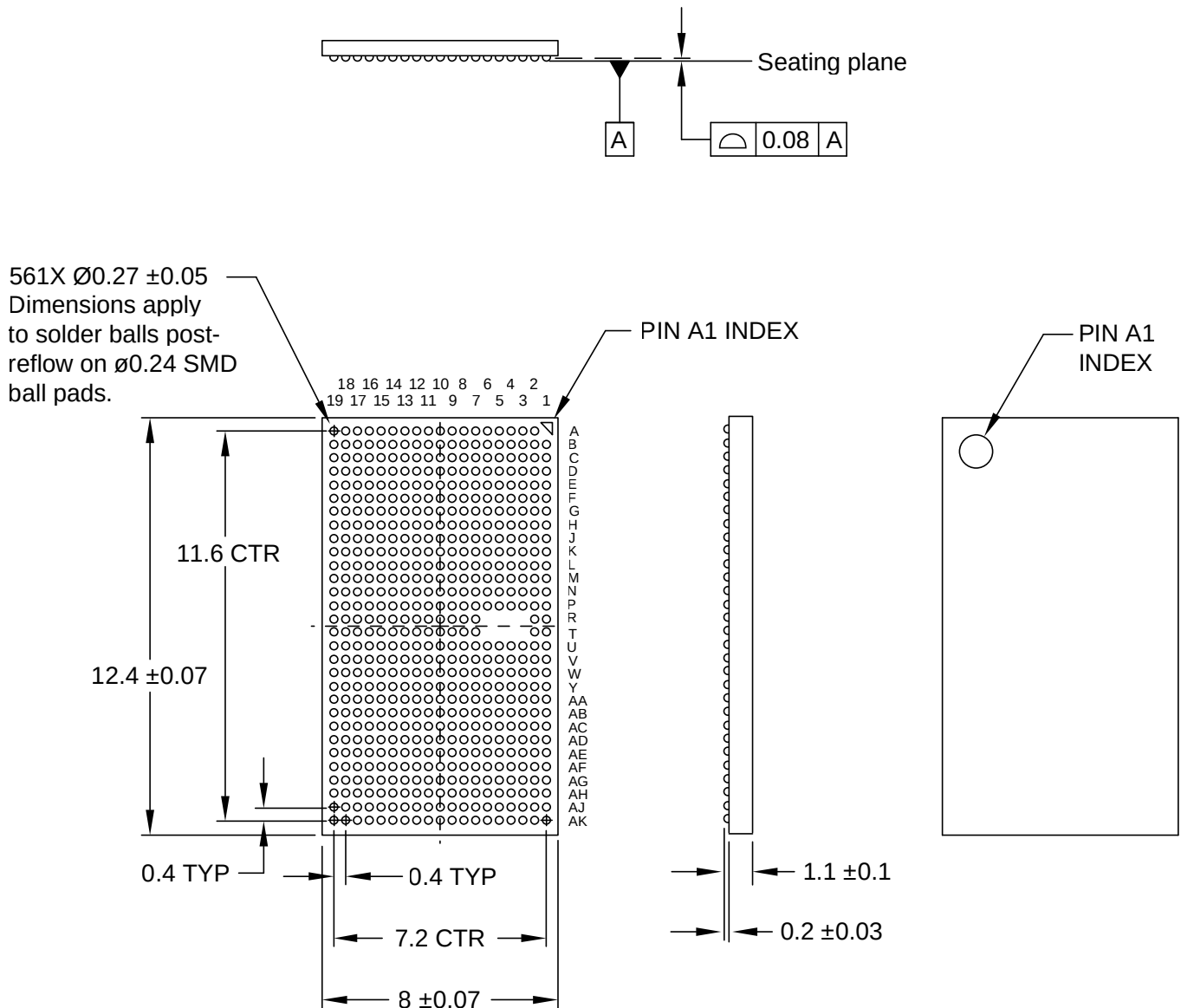


Notes: 1. All dimensions are in millimeters.

2. Solder ball composition: SACQ with CuOSP pads (Sn-4Ag-0.5Cu-3Bi-0.05Ni).

561-Ball Package (Package Code: CZ)

Figure 17: 561-Ball TFBGA – 8.0mm (TYP) × 12.4mm (TYP) × 1.2mm (MAX)



Notes: 1. All dimensions are in millimeters.

2. Solder ball composition: SACQ with CuOSP pads (Sn- 4Ag-0.5Cu-3Bi-0.05Ni)

Package Thermal Impedance

Table 16: Package Thermal Impedance Characteristics

Die Revision	Package	Parameter	Symbol	Value	Unit
Rev D	315-ball, DDP, package code DS	Junction-to-case (TOP)	Θ_{JC}	3.3	°C/W
		Junction-to-board	Θ_{JB}	10.1	°C/W
	315-ball, QDP, package code DS	Junction-to-case (TOP)	Θ_{JC}	2.5	°C/W
		Junction-to-board	Θ_{JB}	8.3	°C/W
	315-ball, 8DP, package code DV	Junction-to-case (TOP)	Θ_{JC}	2.4	°C/W
		Junction-to-board	Θ_{JB}	10.0	°C/W
	441-ball, QDP, package code EK	Junction-to-case (TOP)	Θ_{JC}	2.1	°C/W
		Junction-to-board	Θ_{JB}	4.1	°C/W
	441-ball, 8DP, package code EK	Junction-to-case (TOP)	Θ_{JC}	2.0	°C/W
		Junction-to-board	Θ_{JB}	4.3	°C/W
	561-ball, QDP, package code CZ	Junction-to-case (TOP)	Θ_{JC}	7.2	°C/W
		Junction-to-board	Θ_{JB}	12.7	°C/W
	561-ball, 8DP, package code CZ	Junction-to-case (TOP)	Θ_{JC}	5.8	°C/W
		Junction-to-board	Θ_{JB}	17.7	°C/W

Product-Specific Mode Register Definition

Table 17: Mode Register Contents

Mode Register	OP7	OP6	OP5	OP4	OP3	OP2	OP1	OP0
MR0	Per-pin DFE	Pre Emphasis	Unified NT ODT behavior mode	DMI output behavior mode	Optimized refresh mode	Enhanced WCK always-on mode	Latency mode	NT ODT timing mode
	OP[0] = 1b: Device supports different NT ODT latency for DQ and RDQS							
	OP[1] = 0b: Device supports x16 mode latency for 1G32, 2G32, 1G64, 2G64 OP[1] = 1b: Device supports x8 mode latency for 4G32							
	OP[2] = 1b: Device supports enhanced WCK always-on mode							
	OP[3] = 1b: Device supports optimized refresh mode							
	OP[4] = 1b: Device supports both DMI behavior mode 1 and 2 and mode selection							
	OP[5] = 1b: The NT ODT behavior follows the unified NT ODT behavior							
	OP[6] = 1b: Device supports Pre Emphasis mode							
	OP[7] = 1b: Device supports Per Pin DFE							
MR1						DRFM support ³	ARFM support ³	CS ODT OP support
	OP[0] = 1b: Device supports CS ODT behavior OP							
	OP[1] = 1b: Device supports ARFM							
	OP[2] = 1b: Device supports DRFM							
MR3				BK/BG ORG				
	OP[4:3] = 00b: BG, 10b: 16B Mode support							
MR4				Refresh Multiplier (Note 6)				
	OP[4:0]: Refer to General LPDDR5/LPDDR5X Specification 1 for mode register definitions.							
MR5	Manufacturer ID							
	1111 1111b: Micron							
MR6	Revision ID1							
	0010 1001b							
MR8	I/O width		Density				Device Type ⁵	
	OP[7:6] = 00b: x16 for 1G32, 2G32, 1G64, 2G64 OP[7:6] = 01b: x8 for 4G32		OP[5:2] = 0110b: 16Gb				OP[1:0] = 11b: LPDDR5X (up to 10.7 Gb/s) 10b: LPDDR5X (up to 9600 Mb/s)	
MR19			WCK2DQ OSC FM					
	OP[5] = 1b: WCK2DQ OSC FM supported							

Table 17: Mode Register Contents (Continued)

Mode Register	OP7	OP6	OP5	OP4	OP3	OP2	OP1	OP0
MR21	WXS				ODTD-CSFS	WXFS	RDCFS	WDCFS
	OP[0] = 1b: WRITE DATA COPY function supported							
	OP[1] = 1b: READ DATA COPY function supported							
	OP[2] = 1b: WRITE X function supported							
	OP[3] = 1b: Device ODTD-CS is supported							
	OP[7] = 1b: Data to be written can be selected with 0 and 1							
MR24	DFES				Read DCA			
	OP[3] = 1b: Device supports Read DCA							
	OP[7] = 1b: Device supports DFE (See Note 4)							
MR26		RDQSTFS						
	OP[6] = 1b: Read/write-based RDQS_t TRAINING function supported							
MR27	RAAMULT				RAAIMT			RFM
	OP[0] = 1b: RFM is required							
	OP[5:1] = 01000b: 64							
	OP[7:6] = 01b: 4X							
MR41					E-DVFSC ODT	DVFSC/E-DVFSC Support		
	OP[2:1] = 01b: Only Enhanced DVFSC Mode supported							
	OP[3] = 0b: Enhanced DVFSC ODT option not supported							
MR43		SBEC rule						
	OP[6] = 1b: Simultaneous SBE on each DQ byte and DMI are independently counted							
MR57					RFMSB	RAADEC		
	OP[1:0] = 10b: 2 × RAAIMT							
	OP[3:2] = 01b: Single Bank Mode supported							
MR75								BRCS
	OP[0] = 1b: BRC is supported							
MR74:MR63, MR164:MR76	Reserved MR bits MR63 through MR74 and MR76 through MR164 are RFU by JEDEC standard and should not be accessed by user.							

- Notes: 1. The contents of mode registers described here reflect information specific to each die in these packages.
2. Refer to General LPDDR5/LPDDR5X Specification 1 for mode registers not described here.
3. Refer to General LPDDR5/LPDDR5X Specification 3 for feature description not described here.
4. Device supports 7-step DFE.
5. Device type only supports Bank Group and 16B modes.
6. MR4:OP[4:0] output from each die in a defined package shall be used to determine the required package level refresh rate. Any other method used to determine a valid refresh rate, places ownership and responsibility of the correct refresh rate on the system developer or end user. TOPER (case surface temperature) shall be used to determine whether operating temperature requirements are being met. Any other method used to determine operating temperature, places ownership and responsibility of the correct operating temperature on the system developer or end user.

I_{DD} Parameters

Refer to I_{DD} Specification Parameters and Test Conditions section in General LPDDR5/LPDDR5X Specifications 2 for detailed conditions.

Table 18: WT I_{DD} Parameters – Single Die

Symbol	Supply	Speed Grade (Mb/s)				Unit	Note
		x8 Mode		x16 Mode			
		9600	10,667	9600	10,667		
I _{DD01}	V _{DD1}	3.70	3.70	3.70	3.70	mA	
I _{DD02H}	V _{DD2H}	49.00	49.00	49.00	49.00		
I _{DD02L}	V _{DD2L}	0.20	0.20	0.20	0.20		
I _{DD0Q}	V _{DDQ}	0.60	0.60	0.60	0.60		
I _{DD2P1}	V _{DD1}	1.30	1.30	1.30	1.30	mA	
I _{DD2P2H}	V _{DD2H}	3.60	3.60	3.60	3.60		
I _{DD2P2L}	V _{DD2L}	0.20	0.20	0.20	0.20		
I _{DD2PQ}	V _{DDQ}	0.60	0.60	0.60	0.60		
I _{DD2PS1}	V _{DD1}	1.30	1.30	1.30	1.30	mA	
I _{DD2PS2H}	V _{DD2H}	3.60	3.60	3.60	3.60		
I _{DD2PS2L}	V _{DD2L}	0.20	0.20	0.20	0.20		
I _{DD2PSQ}	V _{DDQ}	0.60	0.60	0.60	0.60		
I _{DD2N1}	V _{DD1}	1.30	1.30	1.30	1.30	mA	
I _{DD2N2H}	V _{DD2H}	31.50	31.50	32.00	32.00		
I _{DD2N2L}	V _{DD2L}	0.20	0.20	0.20	0.20		
I _{DD2NQ}	V _{DDQ}	0.60	0.60	0.60	0.60		
I _{DD2NS1}	V _{DD1}	1.30	1.30	1.30	1.30	mA	
I _{DD2NS2H}	V _{DD2H}	31.50	31.50	32.00	32.00		
I _{DD2NS2L}	V _{DD2L}	0.20	0.20	0.20	0.20		
I _{DD2NSQ}	V _{DDQ}	0.60	0.60	0.60	0.60		
I _{DD3P1}	V _{DD1}	1.90	1.90	1.90	1.90	mA	
I _{DD3P2H}	V _{DD2H}	15.00	15.00	15.00	15.00		
I _{DD3P2L}	V _{DD2L}	0.20	0.20	0.20	0.20		
I _{DD3PQ}	V _{DDQ}	0.60	0.60	0.60	0.60		
I _{DD3PS1}	V _{DD1}	1.90	1.90	1.90	1.90	mA	
I _{DD3PS2H}	V _{DD2H}	15.00	15.00	15.00	15.00		
I _{DD3PS2L}	V _{DD2L}	0.20	0.20	0.20	0.20		
I _{DD3PSQ}	V _{DDQ}	0.60	0.60	0.60	0.60		
I _{DD3N1}	V _{DD1}	2.70	2.70	2.70	2.70	mA	
I _{DD3N2H}	V _{DD2H}	39.50	39.50	40.00	40.00		
I _{DD3N2L}	V _{DD2L}	0.20	0.20	0.20	0.20		
I _{DD3NQ}	V _{DDQ}	0.60	0.60	0.60	0.60		

Table 18: WT I_{DD} Parameters – Single Die (Continued)

Symbol	Supply	Speed Grade (Mb/s)				Unit	Note
		x8 Mode		x16 Mode			
		9600	10,667	9600	10,667		
I _{DD4R1}	V _{DD1}	20.00	21.00	22.00	23.00	mA	3, 4
I _{DD4R2H}	V _{DD2H}	245.00	260.00	365.00	390.00		
I _{DD4R2L}	V _{DD2L}	0.20	0.20	0.20	0.20		
I _{DD4RQ}	V _{DDQ}	69.00	74.00	138.00	147.00		
I _{DD4W1}	V _{DD1}	18.00	19.00	20.00	21.00	mA	3
I _{DD4W2H}	V _{DD2H}	190.00	200.00	280.00	300.00		
I _{DD4W2L}	V _{DD2L}	0.20	0.20	0.20	0.20		
I _{DD4WQ}	V _{DDQ}	0.60	0.60	0.60	0.60		
I _{DD51}	V _{DD1}	20.00	20.00	20.00	20.00	mA	
I _{DD52H}	V _{DD2H}	195.00	195.00	195.00	195.00		
I _{DD52L}	V _{DD2L}	0.20	0.20	0.20	0.20		
I _{DD5Q}	V _{DDQ}	0.60	0.60	0.60	0.60		
I _{DD5AB1}	V _{DD1}	3.30	3.30	3.30	3.30	mA	
I _{DD5AB2H}	V _{DD2H}	43.00	43.00	44.00	44.00		
I _{DD5AB2L}	V _{DD2L}	0.20	0.20	0.20	0.20		
I _{DD5ABQ}	V _{DDQ}	0.60	0.60	0.60	0.60		
I _{DD5PB1}	V _{DD1}	3.30	3.30	3.30	3.30	mA	
I _{DD5PB2H}	V _{DD2H}	43.00	43.00	44.00	44.00		
I _{DD5PB2L}	V _{DD2L}	0.20	0.20	0.20	0.20		
I _{DD5PBQ}	V _{DDQ}	0.60	0.60	0.60	0.60		

- Notes: 1. Published I_{DD} values except I_{DD4RQ} are the maximum I_{DD} values considering the worst-case conditions of process, temperature, and voltage.
2. BG mode enabled. DVFSC and DVFSQ disabled.
3. BL = 16, DBI disabled.
4. I_{DD4RQ} value is reference only. Typical value. Output load = 5pF; R_{ON} = 40 ohms; T_C = 25°C
5. V_{DD1} = 1.70–1.95V; V_{DD2H} = 1.01–1.12V; V_{DD2L} = 0.87–0.97V; V_{DDQ} = 0.47–0.57V; T_C = –25°C to +85°C
6. I_{DD4R2} and I_{DD4W2} have implemented a more stringent pattern than required by JEDEC and therefore these I_{DD} values are slightly higher than if the JEDEC pattern was used.
7. Notes 1, 2, and 5 apply to entire table.

Table 19: WT I_{DD} Parameters – with Enhanced DVFSC enabled - Single Die

Symbol	Supply	Speed Grade (3200 Mb/s)		Unit	Note
		x8 Mode	x16 Mode		
I _{DD4R-DVFSC1}	V _{DD1}	12.00	13.00	mA	3, 4
I _{DD4R-DVFSC2H}	V _{DD2H}	26.00	27.00		
I _{DD4R-DVFSC2L}	V _{DD2L}	85.00	115.00		
I _{DD4R-DVFSCQ}	V _{DDQ}	44.00	88.00		
I _{DD4W-DVFSC1}	V _{DD1}	11.00	12.00	mA	3
I _{DD4W-DVFSC2H}	V _{DD2H}	26.00	27.00		
I _{DD4W-DVFSC2L}	V _{DD2L}	75.00	105.00		
I _{DD4W-DVFSCQ}	V _{DDQ}	0.60	0.60		
I _{DD5ED1}	V _{DD1}	20.00	20.00	mA	
I _{DD5ED2H}	V _{DD2H}	175.00	175.00		
I _{DD5ED2L}	V _{DD2L}	21.00	21.00		
I _{DD5EDQ}	V _{DDQ}	0.60	0.60		
I _{DD5ABED1}	V _{DD1}	3.30	3.30	mA	
I _{DD5ABED2H}	V _{DD2H}	25.00	25.00		
I _{DD5ABED2L}	V _{DD2L}	19.50	20.00		
I _{DD5ABEDQ}	V _{DDQ}	0.60	0.60		
I _{DD5PBED1}	V _{DD1}	3.30	3.30	mA	
I _{DD5PBED2H}	V _{DD2H}	25.00	25.00		
I _{DD5PBED2L}	V _{DD2L}	19.50	20.00		
I _{DD5PBEDQ}	V _{DDQ}	0.60	0.60		

- Notes: 1. Published I_{DD} values except I_{DD4RQ} are the maximum I_{DD} values considering the worst-case conditions of process, temperature, and voltage.
2. Enhanced DVFSC enabled, DVFSQ enabled, 3200 Mb/s, 16B mode
3. BL = 16, DBI disabled.
4. I_{DD4RQ} value is reference only. Typical value. Output load = 5pF; R_{ON} = 40 ohms; T_C = 25°C
5. V_{DD1} = 1.70–1.95V; V_{DD2H} = 1.01–1.12V; V_{DD2L} = 0.87–0.97V; V_{DDQ} = 0.47–0.57V; T_C = –25°C to +85°C
6. I_{DD4R2} and I_{DD4W2} have implemented a more stringent pattern than required by JEDEC and therefore these I_{DD} values are slightly higher than if the JEDEC pattern was used.
7. Notes 1, 2, 5, and 6 apply to entire table.

Table 20: WT Full-Array Power-Down Self Refresh Current – Single Die

Temperature	Symbol	Supply	Value	Unit	Note
25°C	I _{DD61}	V _{DD1}	0.50 (0.20)	mA	1–4
	I _{DD62H}	V _{DD2H}	1.20 (0.50)		
	I _{DD62L}	V _{DD2L}	0.20 (0)		
	I _{DD6Q}	V _{DDQ}	0.60 (0.03)		
	I _{DD6DS1}	V _{DD1}	0.50 (0.19)	mA	1–4
	I _{DD6DS2H}	V _{DD2H}	1.20 (0.49)		
	I _{DD6DS2L}	V _{DD2L}	0.20 (0)		
	I _{DD6DSQ}	V _{DDQ}	0.60 (0.01)		
45°C	I _{DD61}	V _{DD1}	0.27	mA	1–4
	I _{DD62H}	V _{DD2H}	0.92		
	I _{DD62L}	V _{DD2L}	0		
	I _{DD6Q}	V _{DDQ}	0.03		
	I _{DD6DS1}	V _{DD1}	0.25	mA	1–4
	I _{DD6DS2H}	V _{DD2H}	0.92		
	I _{DD6DS2L}	V _{DD2L}	0		
	I _{DD6DSQ}	V _{DDQ}	0.01		
65°C	I _{DD61}	V _{DD1}	0.48	mA	1–4
	I _{DD62H}	V _{DD2H}	2.30		
	I _{DD62L}	V _{DD2L}	0		
	I _{DD6Q}	V _{DDQ}	0.03		
	I _{DD6DS1}	V _{DD1}	0.46	mA	1–4
	I _{DD6DS2H}	V _{DD2H}	2.30		
	I _{DD6DS2L}	V _{DD2L}	0		
	I _{DD6DSQ}	V _{DDQ}	0.01		
85°C	I _{DD61}	V _{DD1}	4.00	mA	1–4
	I _{DD62H}	V _{DD2H}	23.00		
	I _{DD62L}	V _{DD2L}	0.20		
	I _{DD6Q}	V _{DDQ}	0.60		
	I _{DD6DS1}	V _{DD1}	4.00	mA	1–4
	I _{DD6DS2H}	V _{DD2H}	23.00		
	I _{DD6DS2L}	V _{DD2L}	0.20		
	I _{DD6DSQ}	V _{DDQ}	0.60		

- Notes: 1. I_{DD6} 25°C value listed first is the maximum I_{DD} guaranteed value considering the worst-case conditions of process, temperature, and voltage while I_{DD6} 25°C value in parenthesis () is the typical value in the distribution with nominal V_{DD} and a reference-only value. I_{DD6} 45/65°C is the typical value in the distribution with nominal V_{DD} and a reference-only value. I_{DD6} 85°C is the maximum I_{DD} guaranteed value considering the worst-case conditions of process, temperature, and voltage.
2. DVFS and DVFSQ disabled.
3. V_{DD1} = 1.70–1.95V; V_{DD2H} = 1.01–1.12V; V_{DD2L} = 0.87–0.97V; V_{DDQ} = 0.47–0.57V; T_C = –25°C to +85°C
4. While entering and exiting Self-Refresh modes, all defined/used supply rails (V_{DD1}, V_{DD2H}, V_{DD2L}, V_{DDQ}) are required to be at valid levels. After the Self-Refresh with Power down mode entrance commands are completed and t_{ESPD} timing requirement has been satisfied, the V_{DDQ} power rail may be turned off by the controller.

Table 21: IT I_{DD} Parameters – Single Die

Symbol	Supply	Speed Grade (Mb/s)				Unit	Note
		x8 Mode		x16 Mode			
		9600	10,667	9600	10,667		
I _{DD01}	V _{DD1}	4.10	4.10	4.10	4.10	mA	
I _{DD02H}	V _{DD2H}	59.00	59.00	59.00	59.00		
I _{DD02L}	V _{DD2L}	0.20	0.20	0.20	0.20		
I _{DD0Q}	V _{DDQ}	0.60	0.60	0.60	0.60		
I _{DD2P1}	V _{DD1}	1.60	1.60	1.60	1.60	mA	
I _{DD2P2H}	V _{DD2H}	4.80	4.80	4.80	4.80		
I _{DD2P2L}	V _{DD2L}	0.20	0.20	0.20	0.20		
I _{DD2PQ}	V _{DDQ}	0.60	0.60	0.60	0.60		
I _{DD2PS1}	V _{DD1}	1.60	1.60	1.60	1.60	mA	
I _{DD2PS2H}	V _{DD2H}	4.80	4.80	4.80	4.80		
I _{DD2PS2L}	V _{DD2L}	0.20	0.20	0.20	0.20		
I _{DD2PSQ}	V _{DDQ}	0.60	0.60	0.60	0.60		
I _{DD2N1}	V _{DD1}	1.60	1.60	1.60	1.60	mA	
I _{DD2N2H}	V _{DD2H}	42.50	42.50	43.00	43.00		
I _{DD2N2L}	V _{DD2L}	0.20	0.20	0.20	0.20		
I _{DD2NQ}	V _{DDQ}	0.60	0.60	0.60	0.60		
I _{DD2NS1}	V _{DD1}	1.60	1.60	1.60	1.60	mA	
I _{DD2NS2H}	V _{DD2H}	42.50	42.50	43.00	43.00		
I _{DD2NS2L}	V _{DD2L}	0.20	0.20	0.20	0.20		
I _{DD2NSQ}	V _{DDQ}	0.60	0.60	0.60	0.60		
I _{DD3P1}	V _{DD1}	2.20	2.20	2.20	2.20	mA	
I _{DD3P2H}	V _{DD2H}	18.00	18.00	18.00	18.00		
I _{DD3P2L}	V _{DD2L}	0.20	0.20	0.20	0.20		
I _{DD3PQ}	V _{DDQ}	0.60	0.60	0.60	0.60		
I _{DD3PS1}	V _{DD1}	2.20	2.20	2.20	2.20	mA	
I _{DD3PS2H}	V _{DD2H}	18.00	18.00	18.00	18.00		
I _{DD3PS2L}	V _{DD2L}	0.20	0.20	0.20	0.20		
I _{DD3PSQ}	V _{DDQ}	0.60	0.60	0.60	0.60		
I _{DD3N1}	V _{DD1}	3.10	3.10	3.10	3.10	mA	
I _{DD3N2H}	V _{DD2H}	50.50	50.50	51.00	51.00		
I _{DD3N2L}	V _{DD2L}	0.20	0.20	0.20	0.20		
I _{DD3NQ}	V _{DDQ}	0.60	0.60	0.60	0.60		
I _{DD4R1}	V _{DD1}	21.00	22.00	23.00	24.00	mA	3, 4, 6
I _{DD4R2H}	V _{DD2H}	260.00	275.00	380.00	405.00		
I _{DD4R2L}	V _{DD2L}	0.20	0.20	0.20	0.20		
I _{DD4RQ}	V _{DDQ}	69.00	74.00	138.00	147.00		

Table 21: IT I_{DD} Parameters – Single Die (Continued)

Symbol	Supply	Speed Grade (Mb/s)				Unit	Note
		x8 Mode		x16 Mode			
		9600	10,667	9600	10,667		
I _{DD4W1}	V _{DD1}	19.00	20.00	21.00	22.00	mA	3, 6
I _{DD4W2H}	V _{DD2H}	205.00	215.00	295.00	315.00		
I _{DD4W2L}	V _{DD2L}	0.20	0.20	0.20	0.20		
I _{DD4WQ}	V _{DDQ}	0.60	0.60	0.60	0.60		
I _{DD51}	V _{DD1}	20.00	20.00	20.00	20.00	mA	
I _{DD52H}	V _{DD2H}	200.00	200.00	200.00	200.00		
I _{DD52L}	V _{DD2L}	0.20	0.20	0.20	0.20		
I _{DD5Q}	V _{DDQ}	0.60	0.60	0.60	0.60		
I _{DD5AB1}	V _{DD1}	3.30	3.30	3.30	3.30	mA	
I _{DD5AB2H}	V _{DD2H}	49.00	49.00	50.00	50.00		
I _{DD5AB2L}	V _{DD2L}	0.20	0.20	0.20	0.20		
I _{DD5ABQ}	V _{DDQ}	0.60	0.60	0.60	0.60		
I _{DD5PB1}	V _{DD1}	3.30	3.30	3.30	3.30	mA	
I _{DD5PB2H}	V _{DD2H}	49.00	49.00	50.00	50.00		
I _{DD5PB2L}	V _{DD2L}	0.20	0.20	0.20	0.20		
I _{DD5PBQ}	V _{DDQ}	0.60	0.60	0.60	0.60		

- Notes: 1. Published I_{DD} values except I_{DD4RQ} are the maximum I_{DD} values considering the worst-case conditions of process, temperature, and voltage.
2. BG mode enabled. DVFSC, Enhanced DVFSC and DVFSQ disabled.
3. BL = 16, DBI disabled.
4. I_{DD4RQ} value is reference only. Typical value. Output load = 5pF; R_{ON} = 40 ohms; T_C = 25°C
5. V_{DD1} = 1.70–1.95V; V_{DD2H} = 1.01–1.12V; V_{DD2L} = 0.87–0.97V; V_{DDQ} = 0.47–0.57V (DVFSQ disabled)/0.27–0.37V (DVFSQ enabled); T_C = –40°C to +95°C
6. I_{DD4R2} and I_{DD4W2} have implemented a more stringent pattern than required by JEDEC and therefore these I_{DD} values are slightly higher than if the JEDEC pattern was used.
7. Notes 1 and 5 apply to entire table.

Table 22: IT I_{DD} Parameters – with Enhanced DVFS enabled - Single Die

Symbol	Supply	Speed Grade (3200 Mb/s)		Unit	Note
		x8 Mode	x16 Mode		
I _{DD4R-DVFS} C1	V _{DD1}	13.00	14.00	mA	3, 4, 6
I _{DD4R-DVFS} C2H	V _{DD2H}	31.00	32.00		
I _{DD4R-DVFS} C2L	V _{DD2L}	90.00	120.00		
I _{DD4R-DVFS} CQ	V _{DDQ}	44.00	88.00		
I _{DD4W-DVFS} C1	V _{DD1}	12.00	13.00	mA	3, 6
I _{DD4W-DVFS} C2H	V _{DD2H}	31.00	32.00		
I _{DD4W-DVFS} C2L	V _{DD2L}	80.00	110.00		
I _{DD4W-DVFS} CQ	V _{DDQ}	0.60	0.60		
I _{DD5ED} 1	V _{DD1}	20.00	20.00	mA	
I _{DD5ED} 2H	V _{DD2H}	180.00	180.00		
I _{DD5ED} 2L	V _{DD2L}	23.00	23.00		
I _{DD5ED} Q	V _{DDQ}	0.60	0.60		
I _{DD5ABED} 1	V _{DD1}	3.30	3.30	mA	
I _{DD5ABED} 2H	V _{DD2H}	27.00	27.00		
I _{DD5ABED} 2L	V _{DD2L}	21.50	22.00		
I _{DD5ABED} Q	V _{DDQ}	0.60	0.60		
I _{DD5PBED} 1	V _{DD1}	3.30	3.30	mA	
I _{DD5PBED} 2H	V _{DD2H}	27.00	27.00		
I _{DD5PBED} 2L	V _{DD2L}	21.50	22.00		
I _{DD5PBED} Q	V _{DDQ}	0.60	0.60		

- Notes: 1. Published I_{DD} values except I_{DD4RQ} are the maximum I_{DD} values considering the worst-case conditions of process, temperature, and voltage.
2. Enhanced DVFS enabled, DVFSQ enabled, 3200 Mb/s, 16B mode
3. BL = 16, DBI disabled.
4. I_{DD4RQ} value is reference only. Typical value. Output load = 5pF; R_{ON} = 40 ohms; T_C = 25°C
5. V_{DD1} = 1.70–1.95V; V_{DD2H} = 1.01–1.12V; V_{DD2L} = 0.87–0.97V; V_{DDQ} = 0.47–0.57V; T_C = –25°C to +85°C
6. I_{DD4R2} and I_{DD4W2} have implemented a more stringent pattern than required by JEDEC and therefore these I_{DD} values are slightly higher than if the JEDEC pattern was used.
7. Notes 1, 2, and 5 apply to entire table.

Table 23: IT Full-Array Power-Down Self Refresh Current – Single Die

Temperature	Symbol	Supply	Value	Unit	Note
25°C	I _{DD61}	V _{DD1}	0.50 (0.20)	mA	1–4
	I _{DD62H}	V _{DD2H}	1.20 (0.50)		
	I _{DD62L}	V _{DD2L}	0.20 (0)		
	I _{DD6Q}	V _{DDQ}	0.60 (0.03)		
	I _{DD6DS1}	V _{DD1}	0.50 (0.19)	mA	1–4
	I _{DD6DS2H}	V _{DD2H}	1.20 (0.49)		
	I _{DD6DS2L}	V _{DD2L}	0.20 (0)		
	I _{DD6DSQ}	V _{DDQ}	0.60 (0.01)		
45°C	I _{DD61}	V _{DD1}	0.27	mA	1–4
	I _{DD62H}	V _{DD2H}	0.92		
	I _{DD62L}	V _{DD2L}	0		
	I _{DD6Q}	V _{DDQ}	0.03		
	I _{DD6DS1}	V _{DD1}	0.25	mA	1–4
	I _{DD6DS2H}	V _{DD2H}	0.92		
	I _{DD6DS2L}	V _{DD2L}	0		
	I _{DD6DSQ}	V _{DDQ}	0.01		
65°C	I _{DD61}	V _{DD1}	0.48	mA	1–4
	I _{DD62H}	V _{DD2H}	2.30		
	I _{DD62L}	V _{DD2L}	0		
	I _{DD6Q}	V _{DDQ}	0.03		
	I _{DD6DS1}	V _{DD1}	0.46	mA	1–4
	I _{DD6DS2H}	V _{DD2H}	2.30		
	I _{DD6DS2L}	V _{DD2L}	0		
	I _{DD6DSQ}	V _{DDQ}	0.01		
85°C	I _{DD61}	V _{DD1}	4.00	mA	1–4
	I _{DD62H}	V _{DD2H}	23.00		
	I _{DD62L}	V _{DD2L}	0.20		
	I _{DD6Q}	V _{DDQ}	0.60		
	I _{DD6DS1}	V _{DD1}	4.00	mA	1–4
	I _{DD6DS2H}	V _{DD2H}	23.00		
	I _{DD6DS2L}	V _{DD2L}	0.20		
	I _{DD6DSQ}	V _{DDQ}	0.60		

- Notes: 1. I_{DD6} 25°C value listed first is the maximum I_{DD} guaranteed value considering the worst-case conditions of process, temperature, and voltage while I_{DD6} 25°C value in parenthesis () is the typical value in the distribution with nominal V_{DD} and a reference-only value. I_{DD6} 45/65°C is the typical value in the distribution with nominal V_{DD} and a reference-only value. I_{DD6} 85°C is the maximum I_{DD} guaranteed value considering the worst-case conditions of process, temperature, and voltage.
2. DVFS and DVFSQ disabled.
3. V_{DD1} = 1.70–1.95V; V_{DD2H} = 1.01–1.12V; V_{DD2L} = 0.87–0.97V; V_{DDQ} = 0.47–0.57V; T_C = –40°C to +95°C
4. While entering and exiting Self-Refresh modes, all defined/used supply rails (V_{DD1}, V_{DD2H}, V_{DD2L}, V_{DDQ}) are required to be at valid levels. After the Self-Refresh with Power down mode entrance commands are completed and t_{ESPD} timing requirement has been satisfied, the V_{DDQ} power rail may be turned off by the controller.

Revision History

Rev. G – 03/2026

- Moved 441b and 561b packages to production

Rev. F – 02/2026

- Updated legal status to Production
- Added I_{DD} 6 typical values for 25°C
- Updated CS Rx and DQ Rx Mask to include 10.7 Gb/s

Rev. E – 01/2026

- Added I_{DD} 6 values for 45/65°C
- Updated 315b/441b/561b Thermal impedance
- Updated CS Rx and DQ Rx Mask for 9600 Mb/s

Rev. D – 12/2025

- Removed MT62F1G16D1
- Removed 8533 Mb/s MPNs
- Removed 496b and 563b package information
- Added IT I_{DD} values for 9600 and 10,667 Mb/s

Rev. C – 08/2025

- Updated legal status to Preliminary
- Added reference addendum to 10,667 Mb/s
- Added I_{DD} for 9600 and 10,667 Mb/s
- Updated speed grade and part number guide to 10,667, 9600, and 8533 Mb/s
- Updated 561b CZ package tolerance
- Updated 496b package to GZ

Rev. B – 06/2025

- Added 496-Ball GK package

Rev. A – 06/2024

- Initial Advanced data sheet release

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This data sheet contains minimum and maximum limits specified over the power supply and temperature range set forth herein.
 Although considered final, these specifications are subject to change, as further product development and data characterization sometimes occur.